

Extracorporeal Therapies for Acute Kidney Injury and Heart Failure

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Acute kidney injury (AKI) is a common complication of critically ill patients and is most commonly caused by sepsis, hypoperfusion, congestive heart failure, liver failure, or nephrotoxins. AKI is associated with increased morbidity, length of stay, health care costs, and adverse clinical outcomes, including death, development/progression of chronic kidney disease (CKD), and end-stage kidney disease (ESKD). Furthermore, AKI affects distant organs (e.g., heart, lungs, brain, and liver), contributing to their dysfunction during critical illness. Therefore, much effort is aimed at preventing its development and progression. Despite this, approximately 5% to 10% of patients will ultimately require kidney replacement therapy (KRT).¹ This chapter reviews best practices for providing KRT in critically ill patients and those with advanced heart failure (Box 75.1).

ESSENTIALS OF ACUTE KIDNEY REPLACEMENT THERAPY

Mechanisms of Water and Solute Transport

There are four mechanisms by which water and solute are removed with extracorporeal blood purification techniques. Fluid transport across semipermeable membranes is driven by ultrafiltration (UF), whereas solute transport occurs via convection, diffusion, and adsorption (Fig. 75.1A). These mechanisms are used by the blood purification techniques described in this chapter (Fig. 75.2). The necessary formulas for calculating essential functions are presented in Table 75.1.

Ultrafiltration and Convection

UF is the process by which plasma water is pushed through a semipermeable membrane, driven by a hydrostatic pressure gradient. Its rate, Q_{UF} , is determined by the transmembrane pressure (TMP) and the filter's UF coefficient (DK_{UF} , the product of the membrane's K_{UF} and its surface area).² In KRT, the TMP is the pressure gradient between the blood and effluent compartments; it is generated by the blood pump (causing a positive blood compartment pressure) and the effluent pump, which maintains the desired Q_{UF} . The TMP necessary to maintain a determined Q_{UF} increases over time because the membrane gets coated with macromolecules (creating a secondary membrane), decreasing its permeability. When UF is prescribed as a stand-alone procedure, it is called isolated UF or slow continuous UF (SCUF),³ depending on the technique. An additional mechanism generating UF is by osmotic gradients, which is a key mechanism of volume removal during peritoneal dialysis (PD), but it is of limited significance in KRT.

UF also results in solute clearance via solvent drag (convection). Because convection drags any solute that fits through the pores across the membrane, it allows for small- and medium-solute transport. The convective flux of a solute depends on the Q_{UF} , the membrane surface area, the solute concentration in plasma, and its sieving coefficient.² Solute clearance is limited in isolated UF because progressive volume

depletion limits the amount of attainable UF. This complication is avoided by administering replacement fluids at sufficient quantities to allow for the desired Q_{UF} , and consequently solute clearance, a process referred to as *hemofiltration*. The quantity of replacement fluid to be infused equals the total UF volume (UF_{Total}) minus the volume needed to achieve the desired negative balance in the patient, resulting in a net UF (UF_{NET} , or net volume removed from the patient). From a clearance perspective, the fluid is ideally replaced postfilter (postdilution). However, higher Q_{UF} rates lead to hemoconcentration in the filter, which increases the blood viscosity and facilitates premature clotting of the filter. Thus, the filtration fraction should not exceed 30%.⁴ This problem is ameliorated by increasing blood flow rate (Q_b) or administering some or all of the replacement fluid prefilter, thus diluting the blood before it reaches the filter (predilution). The disadvantage of predilution is that it lowers the concentration of the uremic solutes in the blood reaching the hemofilter, thus decreasing net clearance.^{5,6} The magnitude of the decreased clearance varies depending on the Q_b and replacement fluid rate (Q_r).

Diffusion

Diffusion is the net movement of solutes from a region of higher concentration to one of lower concentration. In KRT, the concentration gradient across the filter membrane is created by running dialysate in the effluent compartment. The diffusive flux follows Fick's law of diffusion⁷; it is proportional to the diffusion coefficient of the solute (as approximated using Stokes-Einstein equation) and inversely proportional to the distance between the compartments (membrane thickness and permeability also add resistance). Because the diffusivity coefficient is inversely proportional to a molecule's radius,² this mechanism results in very efficient transport of the fast-moving small solutes, but inefficient transport of the slower-moving larger solutes, even those small enough to fit through the filter pore. Other factors that influence solute clearance include its protein binding and electrical charge. The differences in solute clearance between diffusion and convection are depicted in Fig. 75.1B.

Adsorption

Adsorption is the process that refers to the binding of blood components to the membrane, and is determined by the specific characteristics of each molecule (i.e., dimension, charge, and structure) and each membrane (i.e., porosity, composition, hydrophobicity, surface potential).² It was originally thought that adsorption may contribute substantially to overall clearance of certain substances during KRT. However, the rapid saturation of the membrane binding sites limits its effectiveness. Consequently, specialized adsorption cartridges with higher affinity and capacity for targeted substance have been developed. Their clinical effectiveness remains unproven.

BOX 75.1 Key Clinical Practice Guidelines**General**

Clinical Practice Guidelines, 5th edition: UK Renal Association (2011)
 Kidney Disease: Improving Global Outcomes (KDIGO): Clinical practice guideline for acute kidney injury (2012)
 Acute Kidney Injury: Prevention, detection and management: National Institute for Health and Care Excellence (2013)
 Kidney replacement therapy in adult and pediatric intensive care: Recommendations by an expert panel (2015)
 Acute Disease Quality Initiative (ADQI) 17; Precision CRRT (2016)
 Acute Disease Quality Initiative (ADQI) 22; AKI Quality Improvement (2019)
 Acute Disease Quality Initiative (ADQI) 25; AKI in COVID-19 (2020)
 International Organization for Standardization documents ISO 11663 (Quality of Dialysis Fluid for Haemodialysis and Related Therapies), ISO 13958 (Concentrates for Haemodialysis and Related Therapies), ISO 13959 (Water for Haemodialysis and Related Therapies), and ISO 26722 (Water Treatment Equipment for Haemodialysis Applications and Related Therapies) (www.iso.org)

Vascular Access and Infections

Kidney Disease Outcomes Quality Initiative (KDOQI) Vascular Access Work Group: Clinical practice guidelines for vascular access (2006)
 Locking solutions for hemodialysis catheters; heparin and citrate—a position paper by the American Society of Diagnostic and Interventional Nephrology (2008)
 European Best Practice Guidelines Expert Group on Hemodialysis, European Renal Association. Section V: Chronic Intermittent Haemodialysis and Prevention of Clotting in the Extracorporeal System (2002)
 Parenteral anticoagulants: American College of Chest Physicians Evidence-Based Clinical Practice Guidelines, 8th edition (2008)
 National Evidence-Based Guidelines for Preventing Healthcare-Associated Infections in NHS Hospitals in England (2007)
 Institute for Healthcare Improvement: *How-to Guide: Prevent Central Line-Associated Bloodstream Infections* (2012)
 Centers for Disease Control and Prevention 2016 Update to the 2001 Recommendations for Preventing Transmission of Infections in Chronic Hemodialysis Patients

Who, When, and How of Acute Kidney Replacement Therapy

Initiating Kidney Replacement Therapy

Classic indications for starting KRT include volume overload, metabolic acidosis, or hyperkalemia refractory to medical management, and overt uremia. However, specific thresholds for these indications are empiric and subject to individual preference. Many have argued for early initiation to prevent, rather than treat these complications. Indeed, KRT is commonly initiated to prior to the development of life-threatening complications. This approach has been supported by early observational studies that found a strong association between the severity of volume overload and mortality.⁸⁻¹⁰ However, a causal relationship is not proven, and it remains unknown whether removal of excessive fluid improves outcomes. Any potential benefits of early initiation must be balanced with the potential harmful complications of KRT, including catheter-related infection, hypotension, electrolyte disturbances, suboptimal dosing of antibiotics, nutrition, and so forth.

Recent randomized controlled trials (RCT) have addressed the question of early versus late initiation of KRT (Table 75.2).¹¹⁻¹⁶ Five used Kidney Diseases Improving Global Outcomes (KDIGO) criteria to determine the severity of AKI.¹¹⁻¹⁵ Of these, one found that starting very early (within 8 hours of KDIGO stage 2 AKI) was associated with better

survival.¹¹ However, this was a single center study of mainly cardiac surgical patients and had a fragility index of 3 (a measure of the robustness or fragility of the results of a clinical trial), thus limiting its external validity. Three of the remaining five studies were multicenter RCTs and had a more representative mix of AKI¹²⁻¹⁴; they found no difference in outcomes between early and late start. Because of the lack of benefit in starting early, the AKIKI 2 (Artificial Kidney Initiation in Kidney Injury 2) study examined whether there was any benefit in implementing an even more delayed strategy; they compared a delayed versus very delayed initiation strategy.¹⁵ Although crude mortality was not different between groups, a prespecified multivariate analysis indicated a higher 60-day mortality in the very delayed group. It is important to note that 38% to 50% of patients randomized to the late start groups in the 3 multicenter RCTs never required KRT. This not only suggests that many patients in the early start group would not have required KRT, but also that our ability to predict who will ultimately require KRT is suboptimal. This highlights the need to identify markers that better predict who will require KRT and then establish its optimal timing.

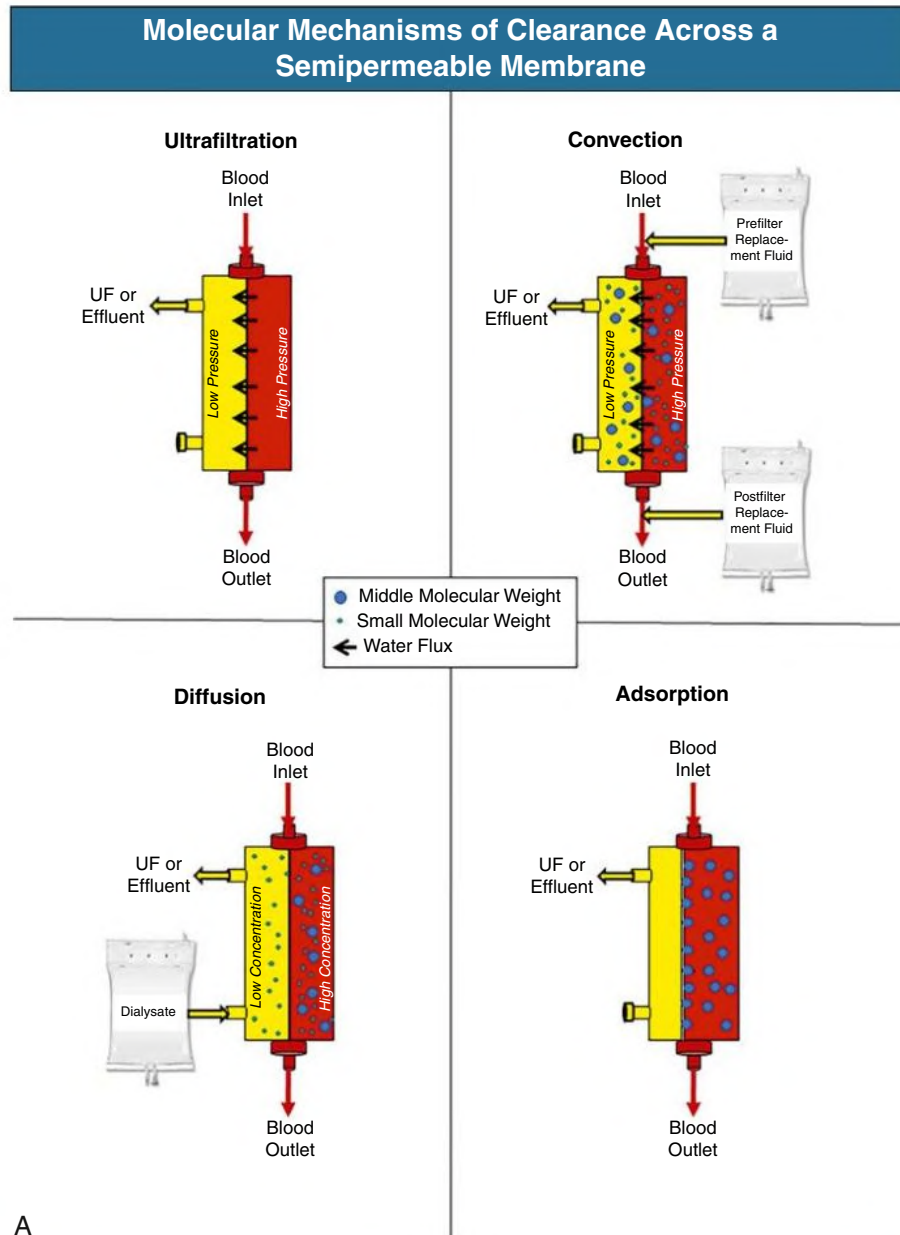
The furosemide stress test* trial was a pilot study that examined this question; they only randomized patients nonresponsive to the furosemide stress test, thus enriching the study population with those patients most likely to require KRT.¹⁵ In this study, 75% of the late patients required KRT, but there was no difference in outcomes between early and late initiation of KRT. The overall evidence obtained from these trials suggests that routinely starting KRT early is not beneficial. However, waiting for absolute indications before considering KRT is also not appropriate and creates logistical challenges. Hence, KRT should be initiated if the burden of fluid and solute derangements is increasing and anticipated to result in complications, remembering that patients with certain illnesses, such as heart and lung disease, may have less tolerance to these homeostatic derangements. A systematic approach to initiating KRT, derived from existing guidelines, is presented in Fig. 75.2.¹⁷

Therapeutic Goals

When preparing the dialysis prescription in acutely ill patients, one should first clearly delineate the goals of therapy and then tailor the prescription to achieve these goals. That is, the prescription should be individualized to the patient. Therapeutic goals for KRT are divided into (1) physiologic targets, (2) dose of KRT delivered, and (3) volume status goals. These goals should be achieved without exacerbating hemodynamic instability, increasing end organ damage, or delaying kidney recovery. *Physiologic targets* refer to the correction of the metabolic abnormalities. In general, ensuring delivery of an adequate dose (see later) is sufficient to normalize most metabolic derangements. However, there are circumstances that may require adjustments of doses and solutions to achieve the desired electrolyte homeostasis (e.g., severe rhabdomyolysis) or clearance of a specific toxin, as may occur during intoxications or severe hyperammonemia (e.g., in infants with inborn errors of urea metabolism, or occasionally postlung transplant patients).[†] Of note, continuous KRT (CKRT) is not an effective therapy for lactic acidosis (hepatic clearance is 10- to 100-fold greater).¹⁸ The only effective treatment is to address the cause of the lactic acidosis (e.g., improving tissue

*The furosemide stress test consists of giving 1 mg/kg intravenous bolus (1.5 mg/kg in patients who are chronically on a loop diuretic). Urine output is then monitored for 2 hours. A urine output of less than 200 mL is considered nonresponsive and predicts progression to renal replacement therapy with an area under the curve of approximately 0.87.

†The use of kidney replacement therapy for this indication has not been established.



Molecular Clearance According to Mechanism and Solute Size

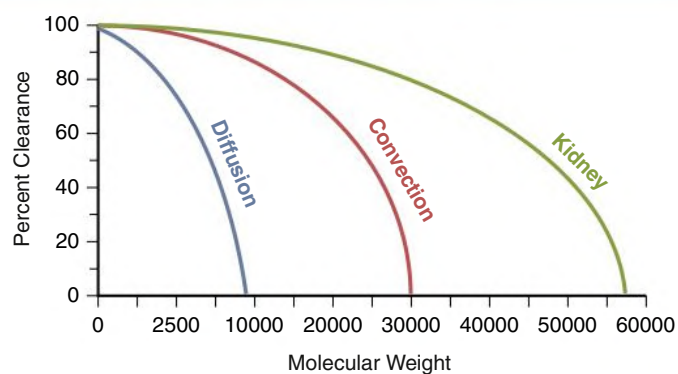


Fig. 75.1 (A) Mechanisms of water and solute clearance in extracorporeal blood purification techniques. (B) A comparison of the size-selective clearance of convection, diffusion, and the native kidney. *UF*, Ultrafiltration.

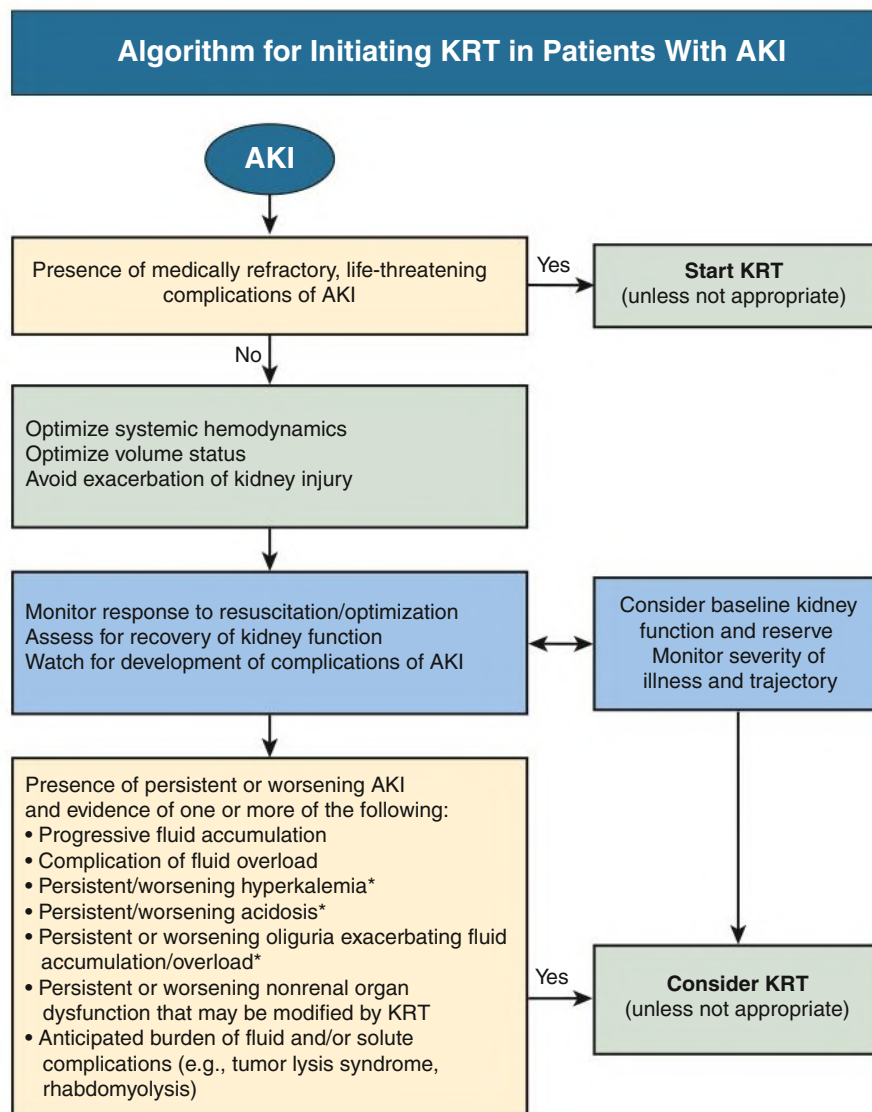


Fig. 75.2 Algorithm for Initiation of Acute Kidney Replacement Therapy (KRT) in Critically ill Patients With Acute Kidney Injury (AKI). *Refractory to medical management. (From Bagshaw SM, Wald R. Strategies for the optimal timing to start renal replacement therapy in critically ill patients with acute kidney injury. *Kidney Int.* 2017;91[5]:1022–1032.)

oxygenation, removing dead gut). Finally, standard modalities are not effective at clearing inflammatory cytokines in a sustained manner and have not been shown to improve clinical outcomes.^{19,20} Thus, cytokine clearance should not be a therapeutic goal of KRT.

The *dose of KRT delivered* should at minimum meet the available evidence-based targets for each modality. In intermittent kidney replacement therapy (IKRT) and prolonged intermittent kidney replacement therapy (PIKRT), adequacy is measured as small solute clearance. The lower limit for intermittent hemodialysis (IHD) is a delivered single-pool Kt/V (spKt/V) of 1.0 per session, because doses below this were associated with decreased survival in patients with intermediate severity of illness.²¹ The VA/NIH ATN study (which permitted switching of KRT modality, depending on the Sequential Organ Failure Assessment [SOFA] cardiovascular score) showed that increasing the delivery of an spKt/V of 1.2 or more, from three to six times weekly, did not provide additional benefit.²² Subsequently, the minimum delivered spKt/V when using IKRT in patients with AKI was recommended to be 1.3 or more per session thrice weekly.²³ However, because solute clearance is commonly decreased in the acutely ill patient, the spKt/V target is frequently not achieved with

standard dialysis prescriptions. Indeed, the ATN study found that delivered spKt/V was approximately 1.1 after the first treatment and the prescribed dialysis time had to be increased to approximately 4 hours. Thus, to avoid underdelivery of IKRT, measurements of the delivered dose should be undertaken and the IKRT prescription adjusted accordingly (Box 75.2). Another way of expressing urea clearance is as continuous small-solute clearance (corrected equivalent renal urea clearance [EKRC]), which kinetically quantifies the “time-averaged KT.” It is particularly useful for PIKRT because EKRC is independent of treatment type and schedule. The Hannover Dialysis Outcome Study of PIKRT found that patients treated with PIKRT to achieve an EKRC of 20 mL/min (plasma urea level = 11.3 ± 4 mmol/L) had outcomes indistinguishable from those in which the EKRC was 13 mL/min (plasma urea levels = 19 ± 6 mmol/L).²⁴ Note that the lower EKRC dose (13 mL/min) in this study was equivalent to a spKt/V of 1.2 to 1.4 per treatment three times weekly. Thus, the recommended EKRC target is a value of 13 mL/min or higher.^{25,26} The nomogram in Fig. 75.3²⁷ expresses the EKRC at distinct combinations of HD dose and frequency, thus providing a comparison between IKRT and PIKRT dosing.

TABLE 75.1 Essential Formulas Used in Kidney Replacement Therapies

Parameter	Formula	Definitions
Transmembrane pressure (TMP)	$\text{TMP} = \frac{(\text{P}_\text{F} + \text{P}_\text{r})}{2} - \text{P}_\text{e}$	P_F = Prefilter pressure P_r = Return pressure P_e = Effluent pressure
Membrane ultrafiltration coefficient (K_{UF})	$\text{K}_{\text{UF}} = \frac{\text{Q}_{\text{UF}}}{\text{TMP}} \times \frac{1}{\text{A}}$	Q_{UF} = Ultrafiltration flow rate A = Membrane surface area
Filter ultrafiltration coefficient (DK_{UF})	$\text{DK}_{\text{UF}} = \text{K}_{\text{UF}} \times \text{A}$	
Ultrafiltration rate (Q_{UF})	$\text{Q}_{\text{UF}} = \text{DK}_{\text{UF}} \times \text{TMP}$	
Sieving coefficient (SC)	$\text{SC} = \frac{\text{CUF}}{(\text{C}_{\text{pi}} + \text{C}_{\text{po}})/2}$	C_{UF} = Solute concentration in UF C_{pi} = Plasma solute concentrations at the inlet C_{po} = Plasma solute concentrations at the outlet
Filtration fraction (FF)	$\text{FF} = \frac{\text{Q}_{\text{UF}}}{\text{Q}_\text{p}}$	Q_p = Plasma flow rate
Convective flux (J_c)	$\text{J}_\text{c} = \frac{\text{Q}_{\text{UF}}}{\text{A}} \times \text{C}_{\text{pi}} \times \text{SC}$	
Solute diffusive flux (J_d)	$\text{J}_\text{d} = -D \left(\frac{dc}{dx} \right)$	D = Diffusivity coefficient dc = Concentration gradient ($dc = \text{C}_1 - \text{C}_2$) dx = Distance between compartments
Diffusivity coefficient (D)	$D = \frac{k\text{BT}}{6\pi\mu R}$	$k\text{B}$ = Boltzmann constant T = Absolute temperature μ = Viscosity of the medium R = Effective radius of the molecules
Rejection coefficient	$\text{RC} = 1 - \text{SC}$	
CVVH postdilution	$\text{K} = \text{Q}_{\text{UF}} \text{SC} \times 60 / \text{body weight}$	
CVVH predilution	$\text{K} = \text{Q}_{\text{UF}} \times \text{SC} \times / [\text{Q}_{\text{BW}} (\text{Q}_{\text{BW}} + \text{Q})] \times 60 / \text{body weight}$	Q_{BW} = Blood water flow rate Q_R = Replacement fluid rate
CVVHD	$\text{K} = \text{Q}_\text{d} + \text{S}_\text{d}$	Q_d = Dialysate rate S_d = Saturation coefficient
CVVHDF	$\text{K} = (\text{Q}_{\text{UF}} + \text{Q}_\text{d}) \times \text{S}_\text{d}$	

CVVH, Continuous venovenous hemofiltration; CVVHD, continuous venovenous hemodialysis; CVVHDF, continuous venovenous hemodiafiltration.

In CKRT the clearance of urea and other small solutes is proportional to the total effluent flow (the sum of the dialysate and Q_{UF} ; Fig. 75.4).²⁸ CKRT dose is thus expressed as effluent volume flow rate per unit of pre-morbid or pre-intensive care unit (ICU) weight (mL/kg/h). Several early small studies suggested that increasing the CKRT dose from 20 mL/kg/h to more than 35 mL/kg/h may increase survival.^{29,30} However, these findings were not replicated in 2 large RCTs. The first, the VA/NIH ATN trial,²² utilized IKRT and CKRT as mentioned before, whereas the second, the Randomised Evaluation of Normal versus Augmented Level (RENAL) Replacement Therapy,³¹ only used CKRT. Based on these studies, there appears to be a dose of CKRT (20–25 mL/kg/h) above which survival becomes dose independent, and thus routine use of higher doses is not warranted. Although a lower dosing limit has not been established, 20 to 25 mL/kg/h is accepted because this was the lowest dose tested in the trials. In clinical practice, interruptions to CKRT due to diagnostic or therapeutic procedures often compromise the dose delivered, thus it is recommended to prescribe 25 to 30 mL/kg/h to ensure adequate dose delivery.

Fluid management is the other essential therapeutic target. Some patients present with fluid overload and simply need its removal. However, fluid management is usually more complex in critically ill patients. Initially, they often require volume resuscitation due to leaky capillaries and hemodynamic compromise. However, care should be given to avoid overzealous fluid administration because it leads to volume overload, increasing the risk for intraabdominal hypertension,

poor wound healing, and impairment of oxygenation; indeed, it is associated with increased mortality.^{8–10} The resuscitation phase is followed by optimization and de-escalation phases where smaller adjustments are made and fluid may need to be removed. Thus, fluid management is a dynamic process that requires frequent assessment of volume status and targets.

The assessment of volume status is difficult because physical signs are generally not informative and surrogates of cardiac filling (e.g., by central venous filling pressures, cardiac or vena cava dimensions) may be inaccurate in critically ill patients. An alternative approach is to test the effect of therapeutic maneuvers, such as fluid challenge on measures of cardiac filling, stroke volume, and blood pressure. Even after fluid status has been assessed, determining the correct therapeutic goal is difficult and optimal strategies to achieve that goal are uncertain. KRT removes fluid from the intravascular compartment, and thus depends on adequate refilling from the interstitial compartment to maintain adequate hemodynamics. Consequently, close monitoring of hemodynamics is needed and UF targets individualized so that adequate cardiopulmonary and extracellular volume status is achieved while avoiding hypotension. Observational studies have shown that fluid accumulation of greater than 10% of body weight ($[(\text{Total input} - \text{Total output}) / \text{Patient body weight}] \times 100\%$) is associated with poor outcomes; thus maintaining it below this threshold should be considered.³² However, this approach needs to be tested in prospective clinical trials.

TABLE 75.2 Randomized Control Trials Comparing Early Versus Late Kidney Replacement Therapy Initiation

	ELAIN	AKIKI	IDEAL-ICU	STARTR-AKI	AKIKI-2	FST Trial
Location	Germany Single center <i>n</i> = 231	France Multicenter <i>n</i> = 620	France Multicenter <i>n</i> = 488	Multinational Multicenter <i>n</i> = 2927	France Multicenter <i>n</i> = 278	Thailand Multicenter <i>n</i> = 297
Inclusion criteria	KDIGO Stage 2 + NGAL >150 ng/mL	Stage 3 AKI + ventilator (85%) Pressors (85%) Sepsis (56%)	RIFLE Stage F Septic shock + Pressors (100%)	Stage 2 and 3	Stage 3 AKI Oliguria >72 h or BUN 40–50 mmol/L	AKI (any stage) + clinical ATN; FST = NR
Timing of KRT	Early <8 h post-AKI Late <12 h or no initiation	Early <6 h post-AKI Late BUN >40 mmol/L Oliguria >72 h life-threatening	Early <12 h post-AKI Late 48 h postrandomization if no kidney recovery	Early <12 h post-AKI Late AKI ≥72 h Life threatening	Delayed <12 h post-AKI More delayed KRT postponed 1 day, or BUN >50 mmol/L or life-threatening	Early <6 h post-AKI Standard
% of KRT early vs. late	100% vs. 91%	98% vs. 51%	97% vs. 62%	97% vs. 62%	Delayed (98%) More delayed (79%)	98% vs. 75%
Type of KRT	100% CVVHDF	IHD (55%) CKRT (45%)	IHD (43%) PIKRT/CKRT (57%)	IHD, PIKRT, or CKRT (68%)	Delayed: IHD (60%) CKRT (39%) Both (1%) More delayed: IHD (58%) CKRT 40% Both (3%)	
Mortality early vs. late	60 days: 38.4% vs. 50.4% 90 days: 39.3% vs. 54.75%*	60 days: 48.5% vs. 49.7%	90 days: 58% vs. 54%	90 days: 44% vs. 44%	28 days: 38% vs. 45% 60 days: 44% vs. 55%	28 days: 62.1% vs. 58.3%
Duration of stay in ICU	Not significant: 19 vs. 22 days	Not significant: 13 vs. 13 days	Not significant: 12 vs. 12 days	Lower in Early group	No difference: 18 vs. 16 days	No difference: 12 vs. 13.5 days
Mechanical ventilation days	125 vs. 181 hours	No difference: 7 vs. 6 days	No difference: 2 vs. 3 days	No difference	No difference	No difference: 4 vs. 0.5 days

*Statistically significant.

AKI, Acute kidney injury; BUN, blood urea nitrogen; CKRT, continuous kidney replacement therapy; CVVHDF, continuous venovenous hemodiafiltration; FST, furosemide stress test; ICU, intensive care unit; IHD, intermittent hemodialysis; NGAL, neutrophil gelatinase–associated lipocalin; PIKRT, prolonged intermittent kidney replacement therapy; KRT, kidney replacement therapy.

ACUTE KIDNEY REPLACEMENT THERAPY MODALITIES

The main modalities of KRT are IKRT, PIKRT, CKRT and PD. Within each one of these modalities there are variations according to the mechanism of clearance, access, and duration (Table 75.3). The choice of modality is largely based on institutional (preference, expertise, and available resources) and patient factors (hemodynamic stability, comorbidities), as several RCTs did not suggest an overall superiority of one modality over another.^{33–34} It has also been proposed that patients who are hemodynamically unstable have better organ recovery if they are treated with PIKRT or CKRT.^{35,36} However, appropriately powered RCTs have not been designed to test this directly, thus it remains speculative.

However, this does not mean that the modalities are clinically equivalent. For instance, patients on IKRT often require prolonging their therapy time or switching to CKRT because of progressive hemodynamic instability, poor metabolic control, or inability to meet their

UF targets.^{22,33,34} We consider the different modalities to be complementary therapies with distinct strengths and weaknesses; thus, the choice of modality should be based on clinical context. Indeed, most will transition between modalities as their clinical status changes. For most, the modality choice still depends primarily on the desired rate of solute and fluid removal²³ and on hemodynamic parameters.³⁷ In addition, there are specific scenarios in which a particular modality may be preferred. For instance, IKRT is the preferred treatment when rapid clearance is desired (e.g., intoxications), whereas PIKRT and CKRT are more desirable when solute disequilibrium should be minimized to avoid water influx into tissues (e.g., patients at risk for dialysis disequilibrium, including those with increased intracranial pressure or abdominal compartment syndrome).³⁸ PD is provided in a similar manner to ESKD patients. The spectrum of dialytic modalities used in acutely ill patients and their general characteristics are listed in Fig. 75.5.

Economic factors may also play a role, as CKRT can be more expensive. However, economic considerations vary widely between different

programs because of differences in nursing structures, quality of CKRT delivery, and other associated factors.³⁹ Thus, economic evaluations comparing modalities needs to be conducted locally, with careful consideration of underlying assumptions about costs and outcomes.

BOX 75.2 Measures to Increase Intermittent Hemodialysis Dose

- Maximize hemodialyzer surface area (up to 2–2.2 m²)
- Maximize hemodialyzer porosity (high flux)
- Maximize blood flow rate
 - Maximize internal lumen diameters of hemodialysis catheter
 - Optimize blood flow
 - Ensure positioning of the hemodialysis catheter tip at the caval atrial junction and IVC as appropriate
- Use right internal jugular vein approach, particularly in patients with a BMI of ≥ 27 kg/m²
- Minimize access recirculation
 - Ensure positioning of the hemodialysis catheter tip at the caval atrial junction or IVC as appropriate
- Maximize dialysate flow rate
- Add postdilution replacement fluid to increase convective clearance (hemodiafiltration)
- Optimize anticoagulation to reduce hemodialyzer fiber bundle clotting
- Increase treatment frequency (up to daily)
- Increased treatment duration and/or consider PIKRT or CKRT
- Optimize patient's hemodynamics to reduce compartmental urea sequestration

BMI, Body mass index; CKRT, continuous kidney replacement therapy; IVC, inferior vena cava; PIKRT, prolonged intermittent kidney replacement therapy.

Intermittent Kidney Replacement Therapy

IKRT, most frequently performed as IHD, makes use of the equipment, methods, and nursing staff established for chronic hemodialysis (HD). It provides rapid solute clearance and UF during relatively brief (3–5 hour) treatments. To achieve this, the Q_b and Q_d are set at relatively rapid rates (Q_b is commonly 300–500 mL/min and Q_d 300–800 mL/h). The dialysate can be delivered by either a batch or single-pass system. Water purity is important to avoid back-filtration of bacterial contaminants and other impurities, which can be detrimental to the patient (see [Chapter 99](#)). Reference standards for water purity in the ICU setting are the same as those for the ESKD setting. IKRT is the modality that removes solutes the fastest, thus decreasing the time a patient is exposed to the extracorporeal circuit. In IHD, small solutes are removed very quickly, making it an ideal therapy for conditions such as life-threatening hyperkalemia and intoxications. A theoretical disadvantage of IHD is that middle and larger molecules are not removed efficiently because the main mechanism of clearance is via diffusion. However, this can be improved by adding a component of convection, where Q_{UF} is tailored to the desired clearance and replacement fluid delivered to achieve the desired fluid balance (hemodiafiltration [HDF]). The replacement fluid should be sterile and can be the same as those used during CKRT, or generated online as ultrapure dialysate (sterilized via ultrafilters) and infused directly into the extracorporeal blood circuit. Some suggest sterile dialysate for all acute IHD; however, data are insufficient to support a strong recommendation. Finally, high-flux membranes have also been used to provide greater convective removal of middle molecules. Despite the rationale for increasing middle molecule clearance, there is no evidence suggesting improved outcomes.

Relationship Between Renal Urea Clearance and Single-Pool Kt/V

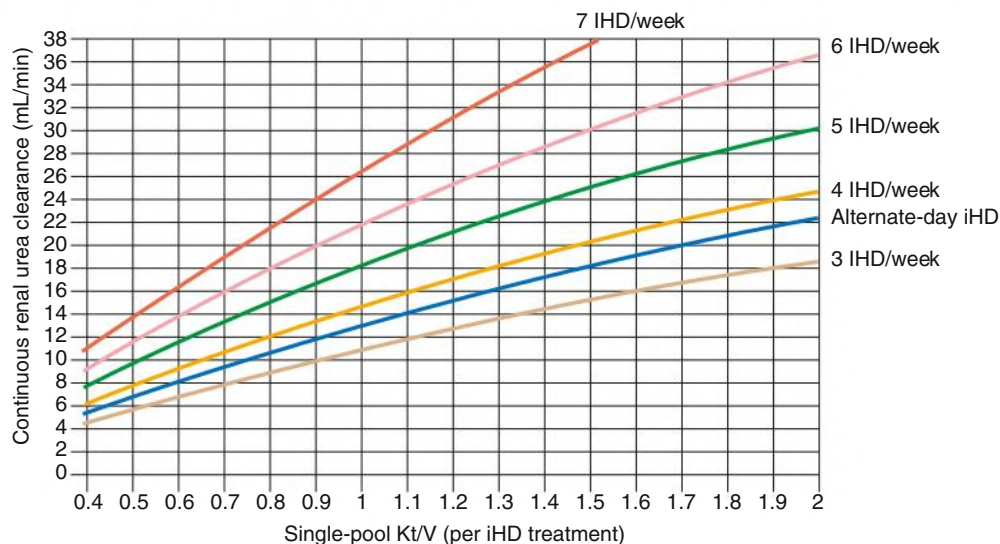


Fig. 75.3 Relationship between corrected continuous kidney urea clearance and single-pool Kt/V (weekly urea clearance/volume of distribution of urea) per treatment for a frequency of three to seven treatments per week. IHD, Intermittent hemodialysis. (From Casino F, Lopez T. The equivalent renal urea clearance: a new parameter to assess dialysis. *Nephrol Dial Transplant*. 1996;11:1574–1581.)

Strategies to Reduce Intradialytic Hemodynamic Instability During Intermittent Kidney Replacement Therapy

The consequences of dialysis-related hypotension are not well studied in the critically ill AKI population but may be detrimental to kidney recovery and end organ function. Indeed, the injured kidney is prone to hypoperfusion due to its impaired autoregulatory responses. Moreover, IKRT can precipitate myocardial stunning, a phenomenon reported in patients with AKI (even treated with CKRT), which in turn may further impair kidney perfusion and impair kidney recovery.⁴⁰ Axiomatically, the most effective measure

to improve hemodynamically stability is increasing the frequency and treatment time, thereby decreasing rates of fluid and solute removal (optimized IKRT). This was one of the main reasons that slower modalities were developed. Studies show better preservation of blood pressure and lower vasopressor requirements in those treated with CKRT and PIKRT compared with IKRT.⁴¹⁻⁴³ Thus, they have become the preferred option for hemodynamically unstable patients.²³ However, switching modalities is not always possible or desired. Consequently, several strategies, borrowed from the ESKD literature, have been employed to avoid hypotension during IKRT in AKI patients. Although bicarbonate-based dialysate is the now the norm, most online dialysate still contains 4 to 8 mmol of acetate, which reportedly may increase blood acetate levels, hypotension, and arrhythmia in certain critically ill patients.⁴⁴ Lowering the dialysate temperature to the lower limits of normal dialysate temperatures (35–36°C) lowers body temperature, which in turn increases vascular resistances and preserves blood pressure during dialysis without apparent deleterious effect.^{45,46} Blood temperature monitoring takes this a step further by controlling thermal transfer to and from dialysate in order to maintain blood temperature at a target value, thus avoiding vasodilation. However, the efficacy of modulating dialysate or blood temperature in the ICU setting has not been firmly established. High dialysate sodium concentrations (145–150 mEq/L) can prevent the fall in effective circulating volume that accompanies the rapid reduction in serum osmolality instigated by IKRT. Sodium profiling further ameliorates this process by inducing water flux into the vascular compartment. A clinical trial showed that intermittent HD with sodium profiling (from 160 to 140 mmol/L) combined with UF profiling improved hemodynamic stability.⁴⁷ Similar data exist for PIKRT. Blood volume monitoring involves biofeedback systems that automatically adjust Q_{UF} and dialysate $[Na^+]$ in response to a fall in circulating blood volume. Although helpful in the ESKD setting, it has not been shown to prevent intradialytic hypotension in the ICU setting.^{48,49} HDF has been suggested to induce less intradialytic hypotension than IHD in the ESKD setting.³⁸ Finally, if the main therapeutic goal is to remove fluid, isolated UF can be performed.³

Prolonged Intermittent Kidney Replacement Therapy

PIKRT was developed as a hybrid between IKRT and CKRT; indeed, urea clearance rates are lower than in IKRT but higher than CKRT. It is usually performed using standard IKRT equipment (CKRT equipment can also be used, as was necessary during the COVID-19 pandemic) but with lower solute clearances and Q_{UF} over longer durations. The intent was to provide the advantages of CKRT (slow, gentle solute clearance and Q_{UF}) while using available IKRT equipment, thus

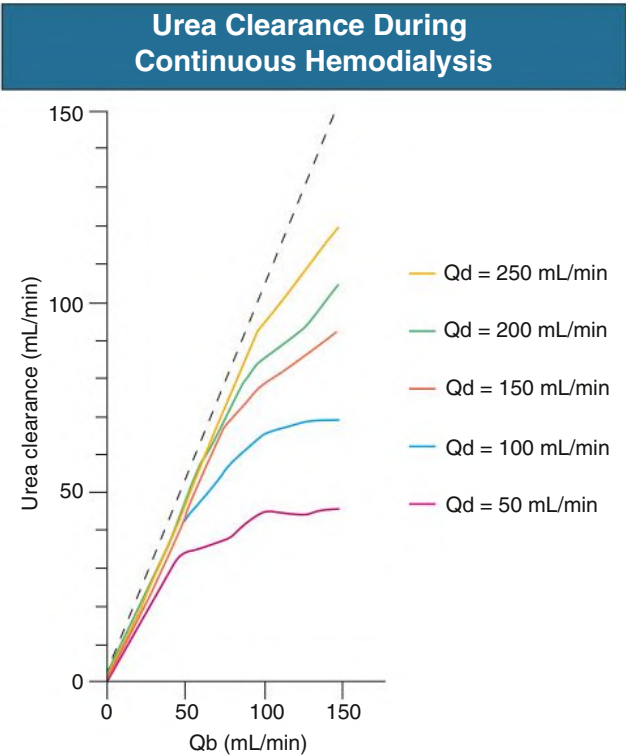


Fig. 75.4 Determinants of Urea Clearance During Continuous Hemodialysis. Relationship among urea clearance, Q_b (blood flow), and Q_d (dialysate flow) during continuous hemodialysis. The flattening of the urea clearance curves describes the conditions in which increases in Q_b do not enhance clearance. (From Kudoh Y, Iimura O. Slow continuous hemodialysis—new therapy for acute renal failure in critically ill patients. Part 1. Theoretical considerations and new technique. *Jpn Circ J*. 1988;52:1171–1182.)

TABLE 75.3 Kidney Replacement Modalities and Their Subtypes

IKRT	PIKRT	CKRT
UF Isolated ultrafiltration (also called sequential ultrafiltration)	EUF Extended ultrafiltration (sometimes called SCUF)	SCUF Slow continuous ultrafiltration
IHD Intermittent hemodialysis	SLED Slow low efficiency dialysis	CVVHD Continuous venovenous hemodialysis
IHF Intermittent hemofiltration	EHF or AVVH Extended hemofiltration or accelerated venovenous Hemofiltration	CVVH Continuous venovenous hemofiltration
IHDF Intermittent hemodiafiltration	EHDF Extended hemodiafiltration	CVVHDF Continuous venovenous hemodiafiltration

CKRT, Continuous kidney replacement therapy; IKRT, intermittent kidney replacement therapy; PIKRT, prolonged intermittent kidney replacement therapy.

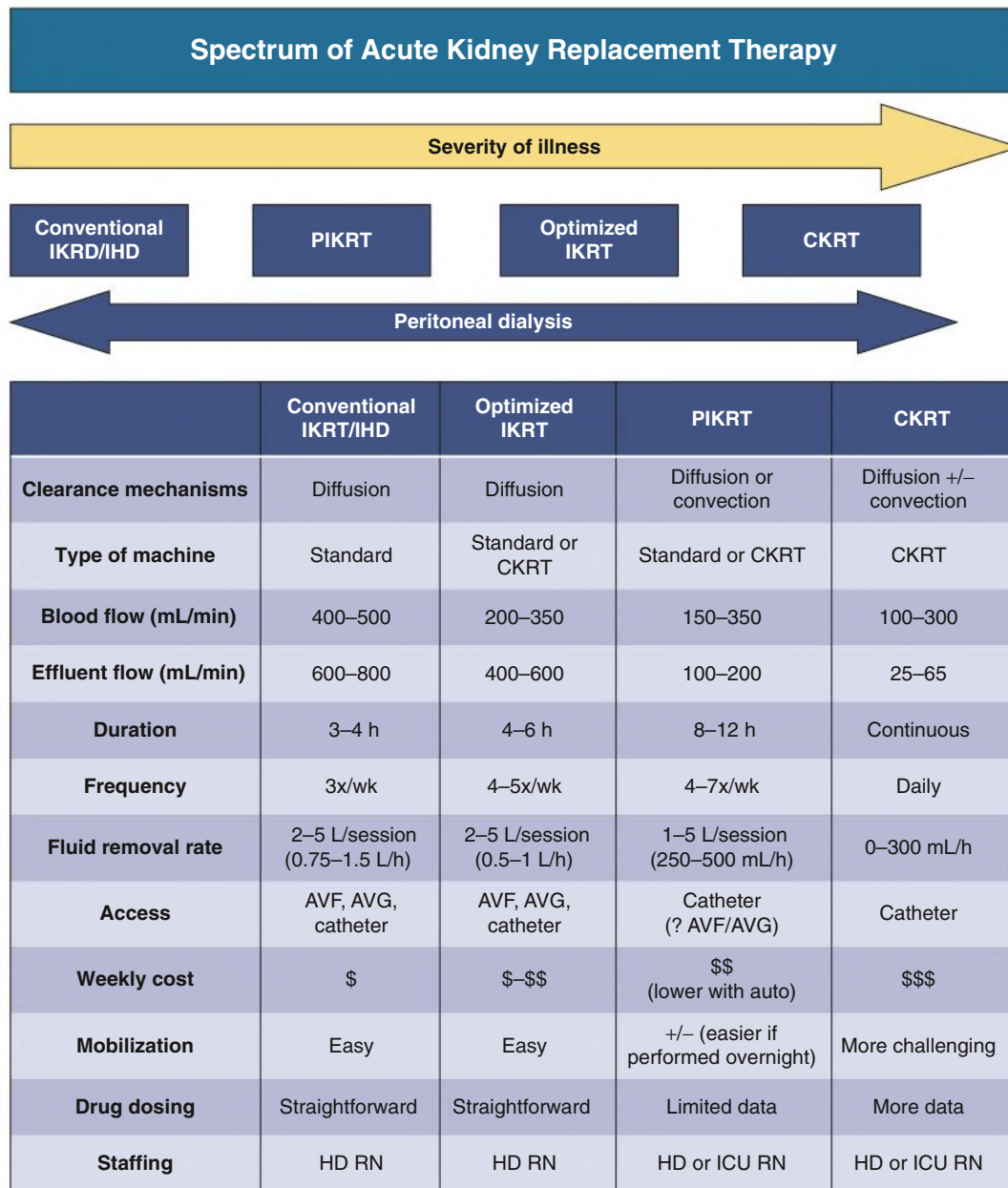


Fig. 75.5 The Spectrum of Kidney Replacement Therapies in Acute Kidney Injury and Their General Characteristics. Although any modality may be used, each has advantages and disadvantages. Thus, the therapy may be tailored to the needs of the individual patient (e.g., patients with increased severity of illness and hypotension may benefit from slower therapies). Optimized IKRT represents IHD prescriptions that are a hybrid between PIKRT and IKRT, which is used to better reach patient therapy goals. AVF, Arteriovenous fistula; AVG, arteriovenous graft; CKRT, continuous kidney replacement therapy; HD, hemodialysis; ICU, intensive care unit; IHD, intermittent hemodialysis; IKRT, intermittent kidney replacement therapy; PIKRT, prolonged intermittent kidney replacement therapy; RN, registered nurse.

minimizing the additional equipment and complexities of implementing a CKRT program.²⁸ PIKRT has the benefit of providing scheduled downtime, which allows the patient to get diagnostic studies, physical therapy, and so forth without compromising dialysis dose. Prescribed correctly, PIKRT provides a high dose of dialysis with minimal urea disequilibrium, excellent control of electrolytes, and good tolerance to UF (Fig. 75.3). PIKRT is usually delivered as a diffusive therapy, although there is increasing experience with combined diffusive and convective clearance. Treatment duration can vary from 6 to 18 hours using a Q_b of 150 to 300 mL/min and Q_d of 50 to 300 mL/min. This wide variation in operating parameters makes it difficult to interpret

the literature on PIKRT and hence precisely define its role. In the older descriptions of PIKRT, the duration (12–18 hours) and other operating characteristics ($Q_b \leq 250$ mL/min, $Q_d \leq 100$ mL/min) were closer to those of CKRT, and treatment tolerance was excellent. The typical duration now is closer to 6 to 8 hours with faster Q_b and Q_d , and thus it may not be as well tolerated in high-risk populations, such as those with septic shock.⁵⁰ For this reason, we prefer lower efficiency prescriptions (i.e., lower Q_{UF} and slower solute removal) in the less stable patients (an algorithm to assist in prescribing PIKRT is presented in Fig. 75.6). Regardless, as with all forms of KRT, quality metrics for the parameters of interest (e.g., Q_{UF} , electrolyte and acid-base

Decision Process for Prescribing Prolonged Intermittent Kidney Replacement Therapy

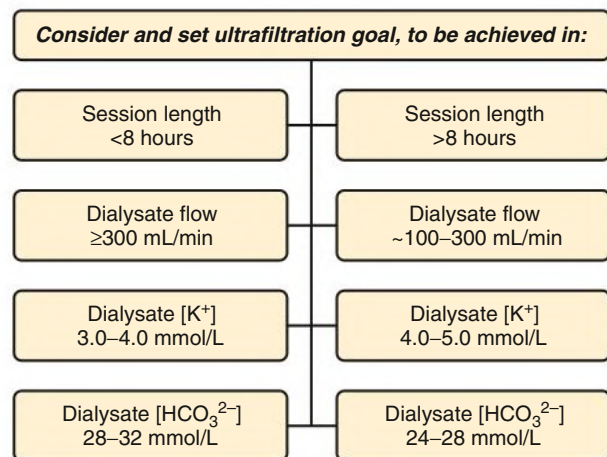


Fig. 75.6 Overview of the decision processes for prescribing prolonged intermittent kidney replacement therapy.

management, anticoagulation, nutrition, medication dosing) should be identified, monitored, and promptly addressed (Table 75.4).

Continuous Kidney Replacement Therapies

CKRT involves the application of low solute clearance and Q_{UF} over 24 hours daily. The lower Q_{UF} provides the best hemodynamic stability of any KRT modality, and the longer treatment duration results in better and more consistent control of uremic solutes, especially for severely catabolic patients. It was first performed in an arteriovenous (AV) configuration in which blood flowed via an arterial catheter, through a simple extracorporeal circuit and filter, and then back to the patient via a venous catheter. This is largely an obsolete practice because of low effectiveness and high complication rate. The technique that is now routinely used is via the venovenous (VV) configuration, where blood is drawn from and returned to the central venous circulation via a mechanical blood pump. CKRT modalities are classified based on their mechanism of molecular clearance (Table 75.3). If the primary mechanism is via convection, it is called continuous venovenous hemofiltration (CVVH); as previously mentioned, the replacement fluid can be returned either prefilter or postfilter. If diffusion is the primary mechanism of clearance, it is called continuous venovenous hemodialysis (CVVHD). A combination of these 2 preceding techniques is called continuous venovenous hemodiafiltration (CVVHDF). The formulas used to calculate small solute clearance were shown in Table 75.1.

The choice of technique depends on whether the primary need is for fluid and/or solute removal, and clinician expertise and preference. For isolated fluid removal (e.g., cardiorenal syndrome), SCUF is used. When solute removal is needed, the physician can choose either the convective or diffusive modalities, as there is clinical equipoise between them.⁵¹ Modern machines are able to provide reliable and rapid Q_b (50–450 mL/min) and dialysate or replacement fluid rates that usually are in the 2 to 4 L/h range, but are capable of running at up to 12 L/h. The major limitations to implementing/sustaining CKRT programs have been cost and complexity. Some contributors to the costs of CKRT include nursing time, filters, and solutions. These are exacerbated by frequent circuit failure (mean operating time for CKRT is reported to be 17–22 h/day), which increases the work load and wastes.⁵² However, implementing simple standardized protocols

can result in the maintenance of a 2:1 patient:nurse ratio and circuit life spans of more than 60 hours, thus significantly decreasing the cost difference between modalities. For this, careful attention must be paid to the technical considerations of CKRT and a continuous quality program implemented.⁵³

Extracorporeal Blood Purification

Critical illness is associated with increased production of cytokines, which are medium and large molecules that modulate the immune response. These trigger an inflammatory state, which is exacerbated in AKI, because cytokine clearance is decreased. Consequently, there has been an ongoing interest developing techniques that reduce cytokines during acute illness, especially sepsis. This interest was accentuated during the COVID-19 pandemic because of the profound inflammation milieu associated with SARS-CoV-2 infection. Cytokines can be removed with high levels of convective clearance, high cutoff hemofilters (membranes with a molecular weight cutoff of 60–150 kDa), bioadsorption devices, and coupled plasma filtration–adsorption (Table 75.5). Note that most of these techniques are not selective: they will remove any circulating substance that meets their clearance characteristics, including proinflammatory and antiinflammatory cytokines, as well as beneficial substances such as medications, including antibiotics. All of these techniques are experimental, supported only by observational studies and pilot trials. Larger multicenter trials are scarce and have not yet provided evidence of benefit. Better techniques are required to identify patients who might benefit from cytokine removal, as well as when and how cytokine removal should be applied.

OPERATIONAL ASPECTS OF ACUTE KIDNEY REPLACEMENT THERAPY

Advances in technology have facilitated the implementation of kidney replacement programs throughout the world. However, ensuring optimal quality and safety remains a challenge. Even now, many institutions lack quality monitoring programs such as monitoring clearance, be it IKRT, PIKRT, or CKRT. The need for defining and standardizing quality metrics in the care of AKI patients was addressed by an Acute Disease Quality Initiative (ADQI) series.⁵³ They emphasize the need for all institutions to implement a quality initiative comprising integrating, monitoring, and reporting of quality metrics for all their KRT therapies. Table 75.4 provides an example of a quality dashboard that can be adapted to fit an institution's needs. It is important to not only meet the individual benchmarks but to also determine the effect they have on overall efficacy and safety of the KRT program. For instance, the reason to monitor catheter function is to prevent premature circuit failure, which not only results in decreased delivery of the KRT dose but also causes blood loss and waste of disposables, alters delivery of medications and nutrition, and increases nurse workload, all of which increase health care costs.

Vascular Access

A prerequisite for IKRT, PIKRT, and CKRT is reliable vascular access through which blood is delivered to the membrane at an adequate flow rate for waste(s) removal but without clotting and circuit dysfunction. If the patient has an AV fistula or graft, this should be used for IKRT. Uncuffed, short-term HD catheters inserted into a central vein are the most commonly used access for CKRT. Although tunneled dialysis catheters may have better delivery metrics and fewer complications, they are usually reserved for patients who either already have one present or those expected to be on KRT for more than 3 weeks. Short-term HD catheters vary in material, size, lumens (number and lumen shape), catheter shape, and tip characteristics.⁵³ However, the

TABLE 75.4 Example of a Quality Metric Dashboard for Kidney Replacement Therapy

Quality Indicator	Operational Definition	Proposed Benchmark
Vascular Access Related		
Access site	Right IJ vs. left IJ vs. femoral vein	For monitoring purposes
Tip placement	Tip within 3 cm of caval-atrial junction (for chest lines only)	1.0
Catheter dysfunction and failure	Number of catheters with dysfunction/total number of catheters	0.8
CLABSIs	CLABSIs/Number of catheter-line days	0 events per catheter
Efficacy: Outcomes		
Kt/V in IKRT and PIKRT	Kt/V after first session (weekly Kt/V to be calculated)	3.9 weekly
Filters used per session	Filters used/Number of sessions	1 per session
Filter life in CKRT	Number of filters lasting 60 h/Total number of filters used	>60% filters
Delivered dose	Actual delivered dose/24 h/Prescribed dose/24 h	>80% of dose
Fluid management	UF removed/24 h/Prescribed UF/24 h	>80% of UF
Small solute clearance	$[sCr(d1) - sCr(d2)]/sCr(d2)$	0
Unplanned time off CKRT	CKRT/24 h	<10% downtime
Downtime	Calculated from machine (e.g., delays in addressing alarms)	<10% downtime
Medication dosing	Appropriate dose adjustments per CKRT	Daily monitoring
Nutrition and supplements	Appropriate nutrition, vitamin, and trace element supplements	Daily monitoring
Safety Indicators		
Adverse events	Number of adverse events/Number of patients on CKRT	0 events
Electrolyte		
Hypophosphatemia	Days with phosphate <2.5/Days of CKRT	0 days
Hypokalemia	Days with potassium <2.5/Days of CKRT	0 days
Hypomagnesemia	Days with magnesium <1.8/Days of CKRT	0 days
Citrate-Related Errors		
Calcium (high or low)	Days with phosphate <2.5/Days of CKRT	0 days
Alkalosis	Days with potassium <2.5/Days of CKRT	0 days
Hypomagnesemia	Days with magnesium <1.8/Days of CKRT	0 days

Structure indicators would be part of the framework of a CKRT quality program and may include a CKRT committee, defined training program, standardized order sets, infrastructure to support and measure these ongoing quality indicators and a process to ensure the proper functioning/maintenance of KRT devices.

CLABSI, Catheter line-associated bloodstream infections; CKRT, continuous kidney replacement therapy; IJ, internal jugular; IKRT, intermittent kidney replacement therapy; PIKRT, prolonged intermittent kidney replacement therapy; UF, ultrafiltration.

TABLE 75.5 Techniques for Extracorporeal Blood Purification of Cytokines and Related Factors

Mechanism of Clearance	Therapy	Remarks
Filtration-based clearance	HVHF	Convective clearance
	High cutoff membranes	Clearance by filtration via larger pore size
	Plasmapheresis	Clearance by plasma exchange
Adsorption-based techniques	AN69ST	Cytokine adsorption
	Cytosorb	Cytokine adsorption
	Depura D2000	Cytokine adsorption
	HA 330	Cytokine adsorption
	oXiris	Endotoxin and cytokine adsorption
	Seraph100	Pathogen adsorption
	Polymyxin B-based adsorption membranes	Adsorption of endotoxin
Combination techniques	CPFA	Combined filtration and adsorption

CPFA, Coupled plasma filtration with adsorption; HA, hemadsorption; HVHF, high-volume hemofiltration.

TABLE 75.6 Determining the Length of Hemodialysis Catheters

Site	Formula	Accuracy
Right internal jugular	Height/10 cm	90%
Left internal jugular	Height/10 + 4 cm	94%
Right subclavian	Height/10 – 2 cm	96%
Left subclavian	Height/10 + 2 cm	97%

most important factors to ensure low resistance/high Q_b are catheter diameter and location of the tip. Thus, the widest catheter with the shortest length that allows for proper tip placement should be used. The catheter length should be chosen based on the insertion site and height using the Peres formulas (Table 75.6).⁵⁴⁻⁵⁶ The right internal jugular has usually been considered to be the first-choice insertion site, followed by the left internal jugular and femoral veins.⁵⁷ The use of the subclavian vein is not recommended, as cannulation can lead to increased risk of venous stenosis. Several studies have challenged the superiority of the internal jugular approach, suggesting that the femoral vein performs as well as the right internal jugular vein (except in the heaviest tertile) and better than the left internal jugular vein.^{58,59} Consequently, the latest guidelines have placed the femoral before the left internal jugular vein. However, it should be noted that the treating center was an independent predictor of short-term HD catheter dysfunction, and it is thus recommended that each center identify their own best practices by monitoring their internal data.⁵³

The demands placed on a short-term HD catheter by and PIKRT are somewhat different than during IKRT (see also Chapter 96). On the one hand, the Q_b demands are less than in IKRT: Q_b usually are 300 mL/min or less. But on the other hand, they are being used from prolonged periods during which it is manipulated or mobilized, thereby increasing the risk of complications and so forth. From a functional standpoint, poorly functioning catheters involve either catheter dysfunction or catheter failure. *Catheter dysfunction* in CKRT is when suboptimal catheter function is causing premature circuit failure or inadequate clearance in the absence of frequent alarms or obvious drops in Q_b , and thus may not be readily detected.⁵⁴ *Catheter failure* is when the Q_b obtainable through the catheter is insufficient to maintain a functioning CKRT; it is characterized by frequent alarms and stopping of the Q_b pump. Causes of early dysfunction or failure include kinking and suboptimal tip placement. Late dysfunction/failure is more often due to intrinsic or extrinsic obstruction caused by thrombosis. Another complication of dialysis catheters is infection. Although largely preventable, they remain problematic, especially in the ICU setting where patients frequently have other intravenous catheters and lines. Hence, catheter-associated bloodstream infections are still too common. There is ongoing controversy around the risk for infection with different insertion sites, but on balance the internal jugular site is preferred, especially in those with a larger body mass index. However, it is likely that the way the catheter is handled is more important at preventing infection than the site in which it is inserted. Low rates of infection require the adherence to specific clinical guidelines and implementing a formal quality improvement program. There is strong evidence that implementing best practice catheter insertion techniques and maintenance results in a close to zero catheter-related infection rate (the core elements are listed in Box 75.3).⁶⁰ The relevant interventional bundles are contained in position statements of the Institute for Healthcare Improvement and the Centers for Disease Control and Prevention Healthcare Infection Control Practices Advisory Committee.^{61,62}

BOX 75.3 Best Practices to Minimize Catheter-Associated Bloodstream Infection**Insertion**

- Hand hygiene and aseptic technique (including early replacement of catheters inserted in uncontrolled settings if adherence to these measures cannot be ensured)
- Maximum barrier precautions (hat, mask, sterile gloves, sterile gown, and full patient drape)
- Appropriate skin preparation using an alcohol-based chlorhexidine (>0.05%) solution; povidone-iodine (preferably with alcohol) or 70% alcohol can be used in patients with chlorhexidine intolerance
- Avoidance of the femoral site for catheter placement, especially in overweight patients ($BMI \geq 27 \text{ kg/m}^2$)
- Avoidance of catheter placement near open wounds

Maintenance

- Daily review of the need for the line, with prompt removal of unnecessary lines
- Appropriate dressing with sterile gauze or a sterile, transparent, semipermeable dressing
- Appropriate schedule for dressing changes according to condition and type of dressing
- Daily review of the catheter exit site with minimal disturbance to the dressing unless clinically indicated
- Appropriate skin preparation before ports are accessed
- Scrub-the-hub protocol for accessing the catheter
- Daily cleansing of patients with a 2% chlorhexidine wash
- Use of a sutureless securing device for catheter stabilization

BMI, Body mass index.

Circuit and Hemofilters

To ensure optimal KRT delivery, a thorough understanding of the circuit and hemofilter is needed. An ideal circuit is characterized by easy priming (low volumes), low resistance to Q_b , biocompatible materials, minimal blood-air interfaces, and easily exchangeable hemofilters (see also Chapter 98). The KRT circuit begins and finishes at the patient's vascular access (Fig. 75.7).⁵⁴ It comprises a series of pumps connected by tubing segments containing sensors that enable the monitoring and regulation of blood and other solutions flowing through the circuit. It has 4 major segments: (1) an inflow (prefilter) segment carrying the blood from the patient's vascular access to the hemofilter, (2) the hemofilter/dialysis membrane, (3) an outflow (postfilter or return segment) carrying "cleansed" blood back to the patient, and (4) the circuits carrying the nonblood solutions. The pressure sensors are strategically located to monitor for specific changes in the circuit dynamics. Any change in pressure detected by a sensor, in the absence of prescription change, reflects a change in the resistance along that segment, thereby facilitating early detection of an abnormality to the circuit dynamics. Modern hemofilters are made of synthetic biocompatible materials with surface areas that vary from as low as 0.1 m² to over 2 m². Although molecular size cutoffs vary between membranes, the primary determinant of size selectivity in standard KRT is the mechanism of clearance (convection vs. diffusion). Although studies have examined the potential benefits of using high cutoff hemofilters to clear more middle molecules (e.g., cytokines), they have not been shown to improve outcomes.

Replacement Fluids and Dialysate

CKRT requires sterile replacement fluid or dialysate for blood purification, with composition that is determined by the clinical requirements

Kidney Replacement Circuit Including Blood Conduit and Nonblood Segments

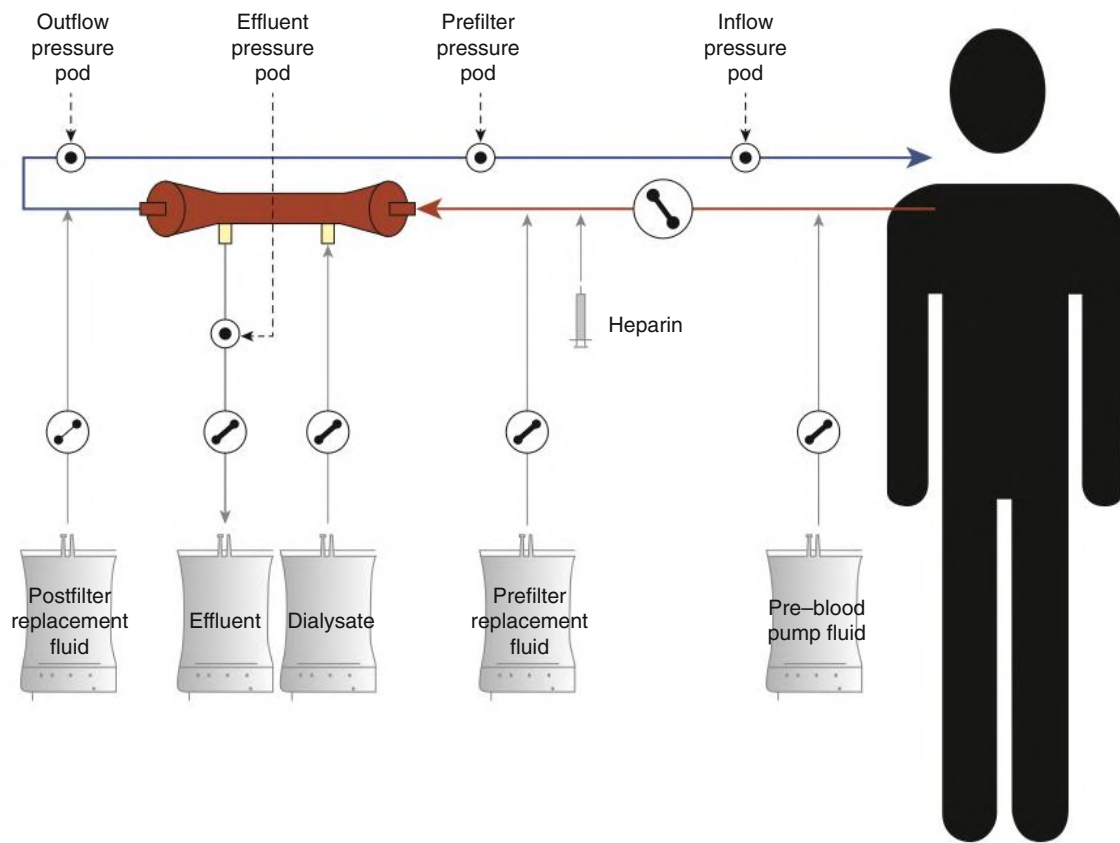


Fig. 75.7 Typical continuous kidney replacement therapy circuit including the blood conduit and nonblood segments. (From Juncos LA, Chandrashekar K, Karakala N, Baldwin I. Vascular access, membranes and circuit for CRRT. *Semin Dial.* 2021;34[6]:406–415.)

for acid-base control and electrolyte management (see also [Chapter 98](#)). Commercially available fluids are the norm, but appropriate fluids can also be prepared derived from peritoneal dialysate and standard intravenous fluids if commercial products are unavailable. Moreover, bagged dialysate solutions for CKRT machines have been generated using single-pass dialysis machines during periods of severe shortages, such as during the COVID-19 pandemic.⁶³ In general, the generation of fluid via pharmacy compounding is avoided because of the increased costs and potential for human error. Bicarbonate buffered solutions are now preferred because they provide superior acid-base control, hemodynamic stability, urea generation, cerebral dysfunction, and may improve survival in select patient groups.⁶⁴ Bicarbonate concentrations in fluid are typically 32 to 35 mmol/L, but 22 to 25 mmol/L concentrations are available for when using citrate anticoagulation. Potassium concentrations vary from 0 to 4 mEq/L. The vast majority of the time a 4 mEq/L solution is sufficient to maintain normokalemia in CKRT, unless the delivered dose is insufficient. The lower potassium concentrations are used when initiating hyperkalemic patients or in those with elevated redistribution of potassium from the intracellular compartment due to tissue breakdown (e.g., rhabdomyolysis). Despite its commercial availability, we do not believe the use of a 0 mEq/L solution is ever warranted in IKRT or PIKRT and is only beneficial in very select/rare circumstances during CKRT. Calcium comes in 2.5, 3, 3.5, or 0 mEq/L configurations. Intravenous calcium replacement is necessary when using the lower calcium concentrations and citrate anticoagulation. Magnesium comes in 1.0 to 1.5 mEq/L configurations, and

therefore periodic repletion of the patient's magnesium levels may be necessary. While phosphate-containing solutions are becoming increasingly available, most continue to lack phosphate; thus supplementation is almost always required after the first few days of CKRT to avoid hypophosphatemia. Phosphate replacement should begin as soon as plasma levels approach low-normal levels and can be accomplished by (1) initiating a continuous infusion (e.g., starting at approximately 1 to 1.5 mmol/h and titrating to maintain normal plasma levels), (2) providing phosphate via the gastrointestinal tract 2 to 3 times daily, or (3) adding appropriate quantities of phosphate into the dialysate.⁶⁵ Because hypophosphatemia has been associated with muscle weakness, including respiratory muscle weakness accompanied by longer duration of mechanical ventilation and mortality, its prevention is an important quality metric to monitor. Finally, dextrose-free solutions may induce euglycemic ketoacidosis; therefore, patients being treated with these solutions should be monitored for this complication.⁶⁵

Anticoagulation in Acute Kidney Replacement Therapy

Most ICU patients can avoid any anticoagulation during intermittent HD because of short treatment duration, but most on PIKRT or CKRT benefit from anticoagulation. Although numerous protocols with anticoagulants have been developed, the most commonly used are based on unfractionated heparin and citrate (see also [Chapter 98](#)).⁶⁶ Unfractionated heparin can be used by infusing heparin either via the syringe in the KRT device, along the inflow segment of the circuit, or systemically. A common protocol starts with an initial bolus dose of 25

to 30 U/kg followed by an infusion of 5 to 20 U/kg/hr titrated to keep the activated partial thromboplastin time (APTT) in the venous blood line 1.5 to 2 times the control value. Advantages of unfractionated heparin include low cost, relative safety, and ease of monitoring. Risks include bleeding in up to 25% to 30% of patients, hyperkalemia, elevated hepatic transaminases, and heparin-induced thrombocytopenia (HIT) in 3% to 5% of patients. To decrease the bleeding risk, a regional heparin/protamine sulfate anticoagulation protocol was proposed in which 1 mg of protamine for every 100 U heparin is infused into the venous line, thus reversing heparin's effect.⁶⁷ However, rebound bleeding can occur when the neutralization with protamine wears off faster than the anticoagulation from heparin. Furthermore, protamine may cause sudden hypotension, bradycardia, or anaphylactoid reactions.

Regional citrate anticoagulation is suggested as the first line strategy by KDIGO guidelines.⁶⁸ It provides the greatest prolongation of filter life and lowest rates of hemorrhage. Regional citrate anticoagulation comprises calcium chelation in the extracorporeal blood circuit and with a calcium infusion either into the return segment or intravenously to restore systemic calcium levels. For intermittent HD and PIKRT, this involves an infusion of 4% trisodium citrate into the arterial blood line, a zero/low-calcium dialysate. For CKRT, either 4% trisodium citrate or anticoagulant citrate dextrose A (ACD-A) is infused into the inflow segment. The rate of citrate is calculated so that blood citrate levels are between 3 and 4 mmol/L or postfilter ionized calcium is less than 0.35 mmol/L. Because the 4% trisodium citrate solution is hypertonic, its use will require an additional infusion of hypotonic fluid or the use of dialysate that is hyponatremic to avoid hyponatremia. Although the ACD-A solution is also hypernatremic, it is not sufficient to induce significant hypernatremia when using appropriate infusion and clearance rates. Both solutions require the use of the lower bicarbonate solutions (22–25 mEq/L) to prevent the alkalosis caused by the subsequent metabolism of the infused citrate into bicarbonate in the liver (in a 1:3 ratio). Frequent monitoring and titration of citrate dose have been advocated to keep the ionized calcium within a therapeutic range. However, many centers now use a simplified fixed-dose anticoagulant citrate dextrose, which minimizes the need to measure postfilter calcium or adjust the citrate infusions. Major complications of regional citrate anticoagulation include systemic hypocalcemia and metabolic alkalosis from citrate toxicity. There is increasing experience with commercial replacement solutions containing citrate as buffer; this chelates calcium in the blood without the need for a separate citrate infusion. Some CKRT devices allow for direct coupling of citrate and calcium infusions, such that a change in Q_b leads to an automatic change in the infusion rates of citrate and calcium. This does not obviate monitoring for systemic hypocalcemia and metabolic alkalosis but is expected to be safer and will likely become the standard approach in the future.

Protocols with other anticoagulants have also been described, but are used less frequently. *Low-molecular-weight heparin* is theoretically advantageous because of increased antithrombotic activity. However, disadvantages include a prolonged half-life (approximately doubled in Acute Kidney Injury Network [AKIN] stage 3, with no significant clearance during KRT), incomplete reversal with protamine, and limited availability of appropriate monitoring by serial anti-factor Xa determinations (recommended level 0.25–0.35 U/mL). Most experience is with dalteparin administered as a single bolus of approximately 20 to 30 U/kg for intermittent HD, followed by an infusion of approximately 10 U/kg/h for PIKRT or CKRT. Overall, the evidence does not support its use over unfractionated heparin in the critically ill AKI setting. *Argatroban* does not cross-react with heparin antibodies and is the preferred approach for HIT because of its hepatic clearance (half-life ~35 minutes, no significant clearance during KRT⁶⁹) and ease of

monitoring with APTT. It is administered as a 0.1 to 0.25 mg/kg bolus before intermittent HD or an infusion of 0.1 to 0.2 mg/kg/h during PIKRT or CKRT, titrated according to APPT. Bivalirudin may be used as an alternative to argatroban in patients with combined kidney and liver failure. Prostacyclin is another effective anticoagulant, but causes variable but occasionally marked falls in BP. Moreover, there is a risk for worsening ventilation-perfusion mismatch and lactic acidosis in patients with multiorgan dysfunction and a risk for increasing intracranial pressure in patients with combined liver and kidney failure. Other less frequently used approaches include antithrombin-dependent factor Xa inhibitors (e.g., fondaparinux) and serine protease inhibitors (nafamostat mesilate).

Drug Dosage in Acute Kidney Replacement Therapy

The effect of AKI and KRT modalities on pharmacokinetics and pharmacodynamic responses remains inadequately studied. Medication dosing data is largely derived from patients with ESKD undergoing IHD and are extrapolated to patients with AKI using the same pharmacologic principals (e.g., size, degree of protein binding, volume of distribution, lipophilicity vs. hydrophilicity). Calculating KRT dosing can be helpful in determining appropriate dosing regimens. For instance, a 70-kg male on CKRT at 25 mL/kg/h (which is 1750 mL/h) corresponds to a glomerular filtration rate (GFR) of approximately 30 mL/min, and the dose of drugs should be calculated accordingly. Drugs with wide therapeutic indices should be used when possible, and drug levels, particularly of those with low therapeutic indices, should be periodically monitored (when possible).

EXTRACORPOREAL KIDNEY THERAPY IN HEART FAILURE AND MECHANICAL CIRCULATORY AND VENTILATORY SUPPORT

Nephrologists are frequently involved in management of patients with heart failure (HF) complicated by diuretic resistance (i.e., failure to reach the therapeutically desired reduction in edema despite adequate and escalating diuretics) and/or kidney dysfunction. The rationale for this is not only because of our expertise in removing fluid (be it by optimizing diuretics or providing KRT) but also because the pathophysiology of HF with diuretic resistance is intimately related to progressive kidney dysfunction (Fig. 75.8). Activation of neurohumoral pathways and venous congestion are of particular importance. The neurohormonal pathways most implicated are the sympathetic nervous system (SNS) and the renin-angiotensin-aldosterone system (RAAS). Activation of these systems induces vasoconstriction of the renal arterioles, alters the autoregulatory curve (shifting it downward and to the right), and promotes sodium retention. Venous congestion exacerbates these maladaptive changes by inducing preglomerular myogenic responses, narrowing the arterial-to-venous pressure gradient, and increasing kidney interstitial pressure, all of which lower kidney perfusion and GFR. Indeed, kidney function in decompensated HF correlates better with central venous pressures than ejection fraction or cardiac index and is associated with increased mortality.⁷⁰ The interplay between the neurohumoral systems and venous congestion is also important when considering mechanisms of decompensation and responses to therapy. That is, because the venous system contains approximately 70% of the circulatory volume, venous vasoconstriction (causing decreased capacitance) can increase preload and decompensate HF without there being an actual increase in body weight.⁷¹ Conversely, increasing venous capacitance can ameliorate HF symptoms despite the absence of fluid removal. This variation in the venous reservoir volume complicates the interpretation of efficacy of fluid removal strategies in HF.

Reciprocal Pathophysiologic Pathways Linking Heart Failure, Kidney Dysfunction, and Congestion in Cardiorenal Syndrome

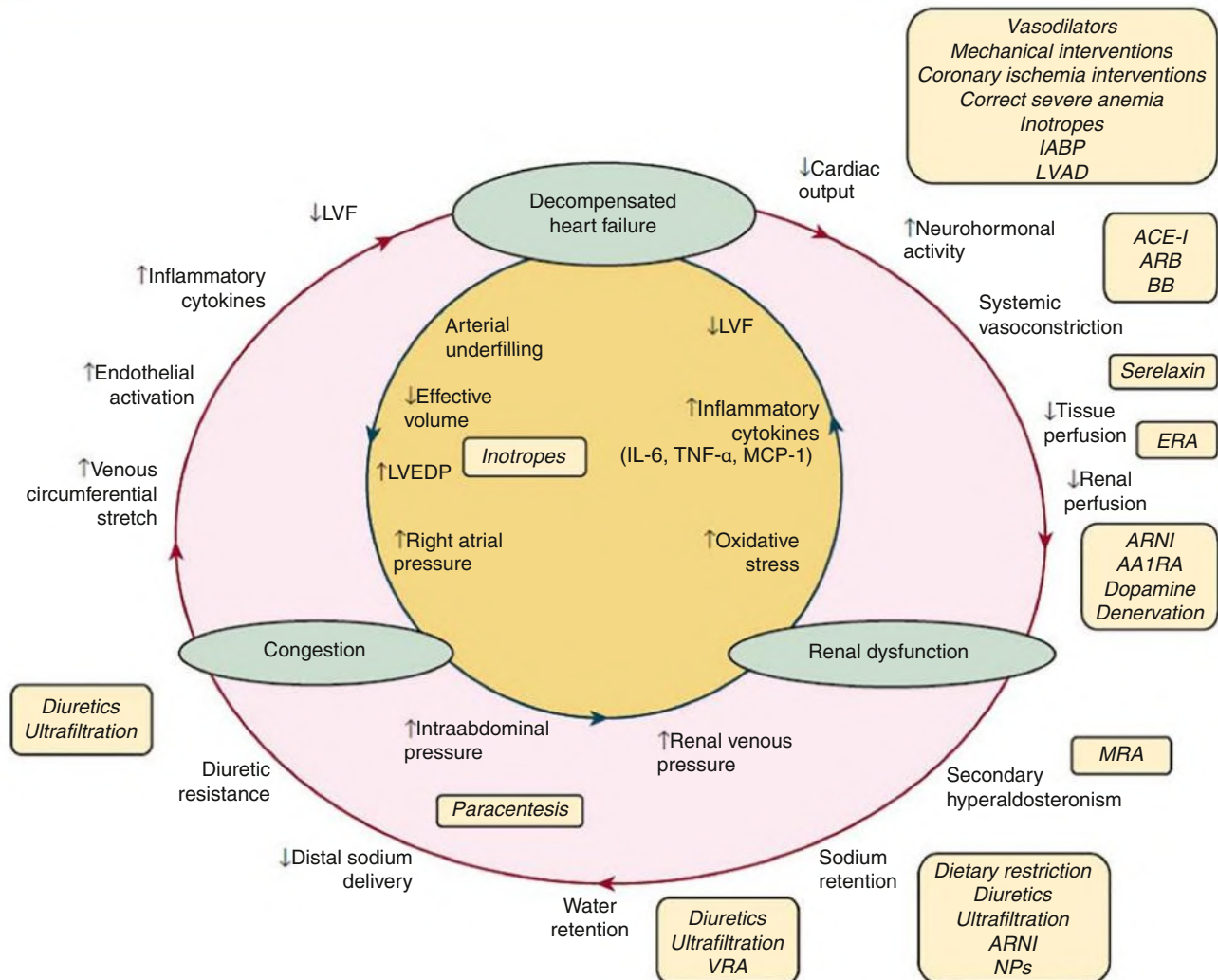


Fig. 75.8 Reciprocal Pathophysiologic Pathways Linking Heart Failure, Kidney Dysfunction, and Congestion in Cardiorenal Syndrome. Decompensation of heart failure can lead to deterioration in kidney function via exacerbated neurohormonal activity and/or kidney venous congestion. The impact of various pharmacologic and nonpharmacologic therapeutic options on the underlying pathophysiologic mechanisms is illustrated. *AA1RA*, Adenosine A1 receptor antagonist; *ACE-I*, angiotensin-converting enzyme inhibitor; *ARB*, angiotensin receptor blocker; *ARNI*, angiotensin receptor neprilysin inhibitor; *BB*, β -blocker; *ERA*, endothelin receptor antagonist; *IABP*, intraaortic balloon pump; *IL-6*, interleukin-6; *LVAD*, left ventricular assist device; *LVEDP*, left ventricular end-diastolic pressure; *LVF*, left ventricular function; *MCP-1*, monocyte chemoattractant protein-1; *MRA*, mineralocorticoid receptor antagonist; *NP*, natriuretic peptide; *TNF- α* , tumor necrosis factor- α ; *VRA*, vasopressin receptor antagonist.

Therapeutic Options for Venous Congestion

Diuretics

Treatment of HF requires addressing treatable causes, instituting diet and lifestyle changes, titrating pharmacologic therapies, and considering mechanical options. The cornerstone of HF therapy for nephrologists revolves around volume management. Indeed, more than 90% of patients admitted with HF are fluid overloaded; the prerenal picture is due to venous congestion, not intravascular volume depletion. Hence, intensifying the diuretic regimen by increasing the dose is the first step. Increasing the frequency or starting a continuous drip are alternative strategies in those with suboptimal responses. RCTs have demonstrated clinical equipoise between approaches, thus one can

tailor it to individual patient responsiveness. A reasonable approach is that used in the DOSE study,⁷² in which the outpatient diuretic dose was increased 2.5-fold and then titrated per clinical responsiveness. Because the dose-response curve to loop diuretics is logarithmic, doubling the dose is warranted until a diuretic response is achieved or the maximal dose reached. In the absence of an adequate response, the reason for resistance should be identified and strategies to improve diuretic responsiveness implemented. Blocking adjacent nephron segments using other diuretics, most commonly thiazide-like agents, may increase natriuresis. Urine sodium monitoring may assist in guiding diuretic regimens. Despite optimization of diuretic regimens, many patients ultimately require other forms of volume removal.

Extracorporeal Ultrafiltration

Intermittent isolated UF and SCUF are characterized by isolated fluid removal; solute clearance is minimal. This was initially reserved for patients with advanced HF and diuretic resistance. Enthusiasm for its use increased greatly following development of commercially available devices that facilitated the procedure and results of early studies.⁷³⁻⁷⁴ Consequently, more definitive studies were attempted. The UNLOAD trial⁷⁵ compared early UF versus intravenous diuretics. Despite the UF group having greater weight loss and fewer rehospitalizations (it was not sufficiently powered to establish this endpoint though), there were no differences in secondary endpoints (dyspnea scores, kidney function, 6-minute walking tests, or mortality). The CARRESS-HF study⁷⁶ did not show superiority of UF to diuretics (using a stepped-based algorithm); it did not improve weight loss, dyspnea, well-being scale scores, or diuretic dose, but had a higher rate of adverse events (kidney failure, bleeding, and catheter-related complications). There were no differences in mortality, emergency visits, or rehospitalizations between groups at 60 days.⁷⁶ These studies led to decreased interest. Indeed, the AVOID-HF trial⁷⁷ was stopped after just 224 (of a planned 810) hospitalized patients were enrolled. Early flexible UF therapy was found to have a greater net fluid loss compared with diuretics without obvious evidence signs of kidney dysfunction. However, the study was underpowered to draw any definitive conclusions regarding hard outcomes. Although several metaanalysis have been performed, the differences between study protocols limit their interpretation. Consequently, isolated UF techniques should now be reserved for patients in whom diuretics are insufficient to achieve/maintain adequate volume status.

Peritoneal Dialysis

PD to treat patients with HF has been used for more than 70 years. When used primarily for UF, PD regimens are simpler than in ESKD; fewer daily exchanges or therapy days per week are needed and thus easily implemented and sustained.⁷⁸ PD catheters can be placed under local anesthesia and used immediately (using low dwell volumes), but laparoscopic insertion, using general anesthesia, is often performed without problems.⁷⁹ Use of PD is associated with improvement in various parameters including electrolytes, fluid status (reduced body weight and diuretic responsiveness), New York Heart Association class, 6-minute walk capacity, hospitalization days, quality of life, cardiac hemodynamics, and HF biomarker levels (BNP, aldosterone); and it does so without activating the SNS or RAAS.⁸⁰ However, this evidence is limited to observational studies and cohorts. Thus, it is possible that the purported benefits of PD may simply reflect patient selection or the extra care provided by the PD/research teams. Available evidence suggests that PD is well tolerated and has a low rate of complications, and thus may be offered as a selective therapy to improve fluid status and symptoms in HF patients with advanced CKD.

Kidney Replacement Therapy in Mechanical Cardiopulmonary Support

The two forms of mechanical circulatory support that are most commonly encountered by nephrologists are extracorporeal membrane oxygenation (ECMO) and ventricular assist devices (VADs). In ECMO, deoxygenated blood is taken from a central vein and pumped through an oxygenator back into either a central vein (VV) or the arterial system (venoarterial), thereby providing respiratory or cardiorespiratory support, respectively. VADs assist the left (LVAD) and right ventricles (RVAD) in patients with refractory heart failure or cardiogenic shock. They are used as a bridge to recovery or transplantation or as destination therapy for patients ineligible for transplantation. Continuous flow devices have almost completely supplanted the earlier pulsatile flow pumps. Another modality of respiratory support available in

some countries is extracorporeal carbon dioxide removal (ECCO₂R). It removes carbon dioxide via a minimally invasive approach that requires lower blood flow and smaller cannulas than ECMO. ECCO₂R can also be achieved, albeit with varying success, via a modified CKRT system linked to a gas exchange membrane.⁸¹

AKI is common in patients with circulatory and/or respiratory failure. If hypoperfusion is causing AKI, initiating mechanical cardiopulmonary support (MCS) may improve kidney function.⁸² However, the incidence of intrinsic AKI (and the need for acute KRT) is high because patients requiring MCS are sicker, and MCS can be complicated by periprocedural hypotension, inflammation, thromboembolism, and hemolysis. Although scant data are available regarding the use of KRT during MCS, the general principles are similar to those for other ICU patients. However, prolonged or continuous acute KRT modalities may be especially advantageous during MCS because they allow for more effective management of patients receiving large fluid volumes. Maintaining adequate volume balance facilitates weaning of MCS, whereas volume overload is associated with worse outcomes.

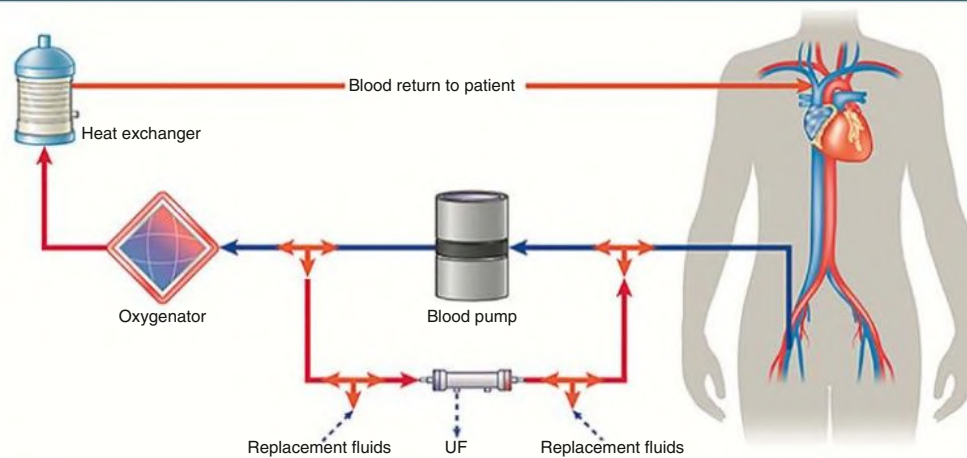
Technical Aspects of Kidney Replacement Therapy During Mechanical Cardiopulmonary Support

The available modalities for providing KRT during MCS are the same as in other situations: PD, IKRT, PIKRT, and CKRT. Preferences vary according to the type of MCS provided. In ECMO, the preferred modality is usually CKRT because it facilitates volume management with less hemodynamic compromise. This is now usually delivered via the standard CKRT devices via HD catheters, or by directly integrating into the ECMO circuit (Fig. 75.9).⁸³ An alternative method, which is simple but less precise, is to provide CKRT by creating a shunt that contains a hemofilter in the ECMO circuit. Blood flow through the shunt is driven by the pressure gradient between these two segments, while infusion pumps regulate dialysate, replacement fluid, and UF rates. PD, which was usually used in infants, is now less common.

The preferred KRT in critically ill patients with VADs, particularly in the postimplantation period, is usually CKRT. This is eventually transitioned to IHD or PD once the patient is hemodynamically stable. Either modality can be continued in the outpatient setting. The most commonly used modality is IHD. This is initially done via a dialysis catheter, but this carries a high risk of infection and should thus be changed over to an AV access as soon as possible. AV grafts are preferred because the fistulas may not mature adequately in the nonpulsatile environment. Another challenge in providing KRT to patients with LVADs is the monitoring and maintenance of adequate preload and afterload during IHD.⁸⁴ This is of vital importance because the function of LVADs depends greatly on these parameters, both of which can change quickly during an IHD session. Afterload is monitored by measuring blood pressure, which is difficult in patients with continuous-flow VADs due to their minimal pulses, thus rendering standard plethysmography unsuitable. Instead, it is measured using Doppler ultrasonography or specialized cuffs with low pulsatility modes.⁸⁴ Stringent blood pressure control is desirable to avoid increased afterload-induced decreases in LVAD pump flow. Preload can be assessed by central venous pressure, echocardiogram, or the pulsatility index (in the Heartmate II). KRT-associated decreases in preload may lead to left ventricular collapse with subsequent fall in MCS flow, pump overdrive, hypotension, and ventricular arrhythmias. This is corrected by gentle volume administration, as other causes of preload reduction (e.g., bleeding, sepsis) are considered. Decreasing the pump speed and treating arrhythmias usually should be performed only after consultation with appropriate VAD personnel.

PD is an underutilized modality in patients with VADs, mainly because of the common misconception that it is contraindicated

In-Line Inclusion of a Hemofilter Into an ECMO Circuit



In-Line Inclusion of a Continuous KRT Device Into an ECMO Circuit

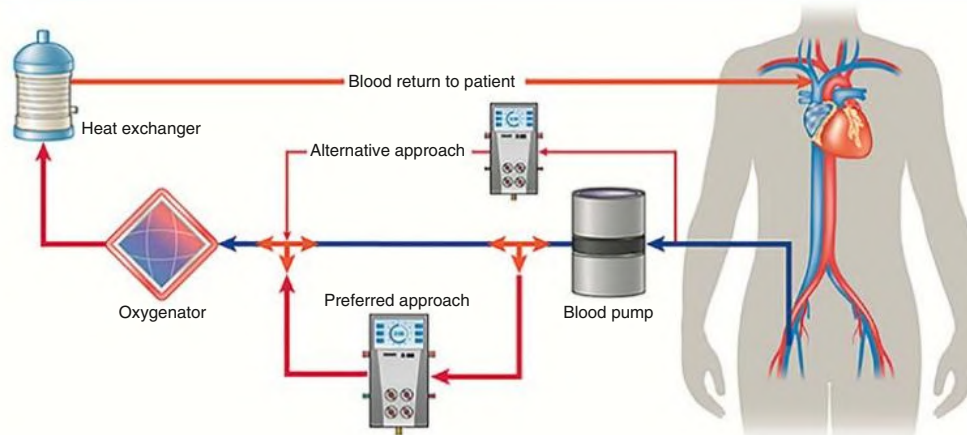


Fig. 75.9 Extracorporeal Membrane Oxygenation (ECMO) Circuit With an In-line Hemofilter (*Top Panel*) or Continuous Kidney Replacement Therapy Device Integrated Into the ECMO Circuit. Two potential connections (preferred and alternative) are illustrated in the bottom panel. *CKRT*, Continuous kidney replacement therapy; *UF*, ultrafiltration. (From Karakala N, Juncos LA. Providing continuous renal replacement therapy in patients on extracorporeal membrane oxygenation. *Clin J Amer Soc Nephrol*. 2020;15[5]:704–706.)

because of increased risk of infection. This concept originated from older generations of VADs, which were implanted into the peritoneal cavity or abdominal wall, and thereby susceptible to infection. However, newer LVADs are extraperitoneal and thus devoid of this intravascular infection risk. Benefits of PD include gentle UF with no abrupt changes in preload, no abrupt electrolyte shifts,

and low risk of bacteremia from catheter infections. Moreover, it is performed at home, which precludes the problem of finding an outpatient dialysis center that will accept them, as most are not comfortable in accepting LVAD patients. More experience is needed to determine the optimal provision of KRT in these patients.

SELF-ASSESSMENT QUESTIONS

- Which of the following statements is *true* with regard to the different acute KRT modalities?
 - The routine use of CKRT rather than IKRT is associated with improved survival.
 - When opting for IKRT, one should routinely use a daily rather than thrice-weekly treatment schedule in order to improve survival.
 - The best available data suggest delivering a minimum dose of 20 mL/kg/h (for CRRT) or a single-pool Kt/V of ≥ 1.2 thrice weekly in critically ill patients receiving hemodialysis.
 - Using higher doses of continuous KRT has been shown to increase cytokine removal and improve outcomes in patients with septic shock and AKI.

2. Which of the following is *true* regarding initiating KRT in critically ill patients?
 - A. A positive test for *TIMP2/IGFBP7* predicts the future development of severe AKI; thus the patient should be prepared for initiation of acute KRT.
 - B. The use of biomarkers of AKI (e.g., neutrophil gelatinase-associated lipocalin) improves diagnosis and facilitates deciding who should receive KRT.
 - C. Using the furosemide stress test to assist in the decision to initiate KRT was found to improve clinically meaningful patient outcomes and prevent the unnecessary initiation of KRT in patients who rapidly recovered renal function.
 - D. The available randomized trials have failed to identify markers that dictate the initiation of KRT in a manner that has a favorable effect on meaningful clinical outcomes.
 3. Which of the following statements is *true* regarding central venous catheters for acute KRT in critically ill patients?
 - A. Temporary nontunneled dialysis catheters placed in the femoral vein have the same catheter dysfunction and infection rates as those placed in the internal jugular veins.
 - B. Tunneled dialysis catheters are associated with better function and fewer infections than temporary nontunneled dialysis catheters in critically ill patients. Thus, evidence-based guidelines recommend that they be used in patients expected to require KRT for more than 7 days.
 - C. Catheters should be placed and used in all critically ill patients requiring KRT, regardless of the presence of arteriovenous fistulas or grafts.
 - D. Internal jugular catheters are routinely preferred to femoral ones.
 4. Which of the following interventions is *not* recommended to reduce central venous catheter-associated bloodstream infection during KRT for critically ill patients?
 - A. A formal quality improvement framework and program for catheter insertion and maintenance
 - B. Cleaning of catheter tips with a chlorhexidine-based solution
 - C. Routine use of antibiotic locks
 - D. Routine placement of catheters in the internal jugular site in obese patients
 5. Which of the following statements is *false* regarding anticoagulation for acute KRT in critically ill patients?
 - A. Patients receiving anticoagulation with heparin have a lower incidence of circuit failure but a higher incidence of bleeding complications compared with to no anticoagulation.
 - B. Argatroban is a preferred approach in those with heparin-induced thrombocytopenia.
 - C. A high calcium gap (citrate lock) is a common adverse event associated with citrate use.
 - D. Periodic flushing of the circuit with saline has been found to prolong circuit life in patients who are not on anticoagulation.
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REFERENCES

- Waikar SS, Curhan GC, Wald R, McCarthy EP, Chertow GM. Declining mortality in patients with acute renal failure, 1988 to 2002. *J Am Soc Nephrol*. 2006;17:1143–1150.
- Neri M, Villa G, Garzotto F, et al. Nomenclature Standardization Initiative: a nomenclature for renal replacement therapy in acute kidney injury: basic principles. *Crit Care*. 2016;20:318.
- Paganini EP, Nakamoto S. Continuous slow ultrafiltration in oliguric acute renal failure. *Am Soc Artif Intern Organs J*. 1980;26.
- Joannidis M, Oudemans-van Straaten HM. Clinical review: patency of the circuit in continuous renal replacement therapy. *Crit Care (London, England)*. 2007;11: 218–218.
- Brunet S, Leblanc M, Geadah D, et al. Diffusive and convective solute clearances during continuous renal replacement therapy at various dialysate and ultrafiltration flow rates. *Am J Kidney Dis*. 1999;34: 486–492.
- Troyanov S, Cardinal J, Geadah D, et al. Solute clearances during continuous venovenous haemofiltration at various ultrafiltration flow rates using Multiflow–100 and HF1000 filters. *Nephrol Dial Transplant*. 2003;18:961–966.
- Ronco C, Ghezzi PM, Brendolan A, Crepaldi C, La Greca G. The haemodialysis system: basic mechanisms of water and solute transport in extracorporeal renal replacement therapies. *Nephrol Dial Transplant*. 1998;13(suppl 6):3–9.
- Payen D, de Pont AC, Sakr Y, Spies C, Reinhart K, Vincent JL. A positive fluid balance is associated with a worse outcome in patients with acute renal failure. *Crit Care*. 2008;12:R74.
- Bouchard J, Soroko SB, Chertow GM, et al. Fluid accumulation, survival and recovery of kidney function in critically ill patients with acute kidney injury. *Kidney Int*. 2009;76:422–427.
- Mitchell KH, Carlom D, Caldwell E, Leary PJ, Himmelfarb J, Hough CL. Volume overload: prevalence, risk factors, and functional outcome in survivors of septic shock. *Ann Am Thorac Soc*. 2015;12:1837–1844.
- Zarbock A, Kellum JA, Schmidt C, et al. Effect of early vs delayed initiation of renal replacement therapy on mortality in critically ill patients with acute kidney injury: the ELAIN randomized clinical trial. *JAMA*. 2016;315:2190–2199.
- Gaudry S, Hajage D, Schortgen F, et al. Initiation strategies for renal-replacement therapy in the intensive care unit. *N Engl J Med*. 2016;375:122–133.
- Barbar SD, Clere-Jehl R, Bourredjem A, et al. Timing of renal-replacement therapy in patients with acute kidney injury and sepsis. *N Engl J Med*. 2018;379:1431–1442.
- Bagshaw SM, Wald R, Adhikari NKJ, et al. Timing of initiation of renal-replacement therapy in acute kidney injury. *N Engl J Med*. 2020;383:240–251.
- Gaudry S, Hajage D, Martin-Lefevre L, et al. Comparison of two delayed strategies for renal replacement therapy initiation for severe acute kidney injury (AKIKI 2): a multicentre, open-label, randomised, controlled trial. *Lancet*. 2021;397:1293–1300.
- Lumlertgul N, Peerapornratana S, Trakarnvanich T, et al. Early versus standard initiation of renal replacement therapy in furosemide stress test non-responsive acute kidney injury patients (the FST trial). *Crit Care*. 2018;22:101.
- Bagshaw SM, Wald R. Strategies for the optimal timing to start renal replacement therapy in critically ill patients with acute kidney injury. *Kidney Int*. 2017;91(5):1022–1032.
- Cheungpasitporn W, Zand L, Dillon JJ, Qian Q, Leung N. Lactate clearance and metabolic aspects of continuous high-volume hemofiltration. *Clin Kidney J*. 2015;8:374–377.
- Joannes-Boyau O, Honore PM, Perez P, et al. High-volume versus standard-volume haemofiltration for septic shock patients with acute kidney injury (IVOIRE study): a multicentre randomized controlled trial. *Intensive Care Med*. 2013;39:1535–1546.
- Mayeur N, Rostaing L, Nogier MB, et al. Kinetics of plasmatic cytokines and cystatin C during and after hemodialysis in septic shock-related acute renal failure. *Crit Care*. 2010;14(3):R115.
- Paganini EP, Tapolyai MB, Goormastic M, et al. Establishing a dialysis therapy/patient outcome link in intensive care unit acute dialysis for patients with acute renal failure. *Am J Kidney Dis*. 1996;28.
- Palevsky PM, Zhang JH, O'Connor TZ, et al. Intensity of renal support in critically ill patients with acute kidney injury. *N Engl J Med*. 2008;359:7–20.
- Dialysis interventions for treatment of AKI. Section 5. *Kidney Int Suppl* (2011). 2012;2:89–115, 2012.
- Faulhaber-Walter R, Hafer C, Jahr N, et al. The Hannover Dialysis Outcome study: comparison of standard versus intensified extended dialysis for treatment of patients with acute kidney injury in the intensive care unit. *Nephrol Dial Transplant*. 2009;24:2179–2186.
- Casino FG, Marshall MR. Simple and accurate quantification of dialysis in acute renal failure patients during either urea non-steady state or treatment with irregular or continuous schedules. *Nephrol Dial Transplant*. 2004;19:1454–1466.
- Claure-Del Granado R, Macedo E, Chertow GM, et al. Toward the optimal dose metric in continuous renal replacement therapy. *Int J Artif Organs*. 2012;35:413–424.
- Casino F, Lopez T. The equivalent renal urea clearance: a new parameter to assess dialysis. *Nephrol Dial Transplant*. 1996;11:1574–1581.
- Kudoh Y, Iimura O. Slow continuous hemodialysis—new therapy for acute renal failure in critically ill patients. Part 1. Theoretical considerations and new technique. *Jpn Circ J*. 1988;52:1171–1182.
- Ronco C, Bellomo R, Homel P, et al. Effects of different doses in continuous veno-venous haemofiltration on outcomes of acute renal failure: a prospective randomised trial. *Lancet*. 2000;356:26–30.
- Saudan P, Niederberger M, De Seigneux S, et al. Adding a dialysis dose to continuous hemofiltration increases survival in patients with acute renal failure. *Kidney Int*. 2006;70:1312–1317.
- The RENAL Replacement Therapy Study Investigators. Intensity of continuous renal-replacement therapy in critically ill patients. *N Engl J Med*. 2009;361:1627–1638.
- Fülöp T, Pathak MB, Schmidt DW, et al. Volume-related weight gain and subsequent mortality in acute renal failure patients treated with continuous renal replacement therapy. *ASAIO J*. 2010;56(4):333–337.
- Vinsonneau C, Camus C, Combes A, et al. Continuous venovenous haemodiafiltration versus intermittent haemodialysis for acute renal failure in patients with multiple-organ dysfunction syndrome: a multicentre randomised trial. *Lancet*. 2006;368:379–385.
- Schefold JC, von Haehling S, Pischowski R, et al. The effect of continuous versus intermittent renal replacement therapy on the outcome of critically ill patients with acute renal failure (CONVINT): a prospective randomized controlled trial. *Crit Care*. 2014;18:R11.
- Bagshaw SM, Berthiaume LR, Delaney A, Bellomo R. Continuous versus intermittent renal replacement therapy for critically ill patients with acute kidney injury: a meta-analysis. *Crit Care Med*. 2008;36:610–617.
- Pannu N, Klarenbach S, Wiebe N, Manns B, Tonelli M. Alberta Kidney Disease Network: renal replacement therapy in patients with acute renal failure: a systematic review. *JAMA*. 2008;299:793–805.
- Cerda J, Ronco C. Modalities of continuous renal replacement therapy: technical and clinical considerations. *Semin Dial*. 2009;22:114–122.
- Marshall MR. Measuring the patient response to dialysis therapy: hemodiafiltration and clinical trials. *Kidney Int*. 2017;91:1279–1282.
- Srisawat N, Lawsin L, Uchino S, et al. Cost of acute renal replacement therapy in the intensive care unit: results from the Beginning and Ending Supportive Therapy for the Kidney (BEST Kidney) study. *Crit Care*. 2010;14:R46.
- Slessarev M, Salerno F, Ball IM, McIntyre CW. Continuous renal replacement therapy is associated with acute cardiac stunning in critically ill patients. *Hemodial Int*. 2019;23:325–332.
- Marshall MR, Golper TA. Low-efficiency acute renal replacement therapy: role in acute kidney injury. *Semin Dial*. 2011;24:142–148.
- Kielstein JT, Kretschmer U, Ernst T, et al. Efficacy and cardiovascular tolerability of extended dialysis in critically ill patients: a randomized controlled study. *Am J Kidney Dis*. 2004;43:342–349.
- Schwenger V, Weigand MA, Hoffmann O, et al. Sustained low efficiency dialysis using a single-pass batch system in acute kidney injury—a

- randomized interventional trial: the REnal Replacement Therapy Study in Intensive Care Unit PatiEnts. *Crit Care*. 2012;16:R140.
44. Unarokov ZM, Mukhoedova TV, Shuvaeva OV. Comparison of sustained low-efficiency dialysis with acetate-free and acetate-containing bicarbonate dialysate in unstable patients: reply to comments. *Artif Organs*. 2014;38:906–907.
 45. Yu AW, Ing TS, Zabaneh RI, Daugirdas JT. Effect of dialysate temperature on central hemodynamics and urea kinetics. *Kidney Int*. 1995;48:237–243.
 46. Jost CM, Agarwal R, Khair-el-Din T, Grayburn PA, Victor RG, Henrich WL. Effects of cooler temperature dialysate on hemodynamic stability in “problem” dialysis patients. *Kidney Int*. 1993;44:606–612.
 47. Paganini EP, Sandy D, Moreno L, Kozlowski L, Sakai K. The effect of sodium and ultrafiltration modelling on plasma volume changes and haemodynamic stability in intensive care patients receiving haemodialysis for acute renal failure: a prospective, stratified, randomized, cross-over study. *Nephrol Dial Transplant*. 1996;11(suppl 8):32–37.
 48. du Cheyron D, Terzi N, Seguin A, et al. Use of online blood volume and blood temperature monitoring during haemodialysis in critically ill patients with acute kidney injury: a single-centre randomized controlled trial. *Nephrol Dial Transplant*. 2013;28:430–437.
 49. Tonelli M, Astephen P, Andreou P, et al. Blood volume monitoring in intermittent hemodialysis for acute renal failure. *Kidney Int*. 2002;62:1075–1080.
 50. Sun Z, Ye H, Shen X, Chao H, Wu X, Yang J. Continuous venovenous hemofiltration versus extended daily hemofiltration in patients with septic acute kidney injury: a retrospective cohort study. *Crit Care*. 2014;18:R70.
 51. Friedrich JO, Wald R, Bagshaw SM, Burns KE, Adhikari NK. Hemofiltration compared to hemodialysis for acute kidney injury: systematic review and meta-analysis. *Crit Care*. 2012;16:R146.
 52. Rewa O, Tolwani A, Ronco C, et al. Quality of care and safety measures of acute renal replacement therapy: workgroup statements from the 22nd Acute Disease Quality Initiative (ADQI) Consensus Conference. *J Crit Care*. 2019;54:52–57.
 53. Kashani K, Rosner MH, Haase M, et al. Quality improvement goals for acute kidney injury. *Clin J Am Soc Nephrol*. 2019;14:941–953.
 54. Juncos LA, Chandrashekar K, Karakala N, Baldwin I. Vascular access, membranes and circuit for CRRT. *Semin Dial*. 2021.
 55. Peres PW. Positioning central venous catheters—a prospective survey. *Anaesth Intensive Care*. 1990;18:536–539.
 56. Czepizak CA, O’Callaghan JM, Venus B. Evaluation of formulas for optimal positioning of central venous catheters. *Chest*. 1995;107:1662–1664.
 57. Oliver MJ. Acute dialysis catheters. *Semin Dial*. 2001;14:432–435.
 58. Baldwin I, Bellomo R, Koch B. Blood flow reductions during continuous renal replacement therapy and circuit life. *Intensive Care Med*. 2004;30(11):2074–2079.
 59. Parienti JJ, Thirion M, Mégarbane B, et al. Femoral vs jugular venous catheterization and risk of nosocomial events in adults requiring acute renal replacement therapy: a randomized controlled trial. *JAMA*. 2008;299(20):2413–2422.
 60. McLaws ML, Burrell AR. Zero risk for central line-associated bloodstream infection: are we there yet? *Crit Care Med*. 2012;40:388–393.
 61. Improvement IH. *How-to Guide: Prevent central Line-Associated Bloodstream Infections*. Cambridge, MA: Institute for Healthcare Improvement; 2012.
 62. O’Grady NP, Alexander M, Burns LA, et al. Guidelines for the prevention of intravascular catheter-related infections. *Clin Infect Dis*. 2011;52:e162–e193.
 63. Moses AA, Stevens JS, Fine D, et al. Protocol for local on-site dialysate production for continuous renal replacement therapy during the COVID-19 pandemic. *Kidney360*. 2021;2:1152–1155.
 64. Kashani K, Thongprayoon C, Cheungpasitporn W, Iacovella GM, Akhoundi A, Albright Jr RC. Association between mortality and replacement solution bicarbonate concentration in continuous renal replacement therapy: a propensity-matched cohort study. *PLoS One*. 2017;12. e0185064–e0185064.
 65. Ebah LM, Akhtar M, Wilde I, et al. Phosphate enrichment of dialysate for use in standard and extended haemodialysis. *Blood Purif*. 2012;34:28–33.
 66. Tolwani AJ, Wille KM. Anticoagulation for continuous renal replacement therapy. *Semin Dial*. 2009;22:141–145.
 67. Morabito S, Guzzo I, Solazzo A, Muzi L, Luciani R, Pierucci A. Continuous renal replacement therapies: anticoagulation in the critically ill at high risk of bleeding. *J Nephrol*. 2003;16:566–571.
 68. Group, KDIGO KAKIW. KDIGO Clinical practice guideline for acute kidney injury. *Kidney Int Suppl*. 2011;2:1–138.
 69. Bambrah RK, Pham DC, Rana F. Argatroban in heparin-induced thrombocytopenia: rationale for use and place in therapy. *Ther Adv Chronic Dis*. 2013;4:302–304.
 70. Khan NA, Ma I, Thompson CR, et al. Kidney function and mortality among patients with left ventricular systolic dysfunction. *J Am Soc Nephrol*. 2006;17:244–253.
 71. Fallick C, Sobotka PA, Dunlap ME. Sympathetically mediated changes in capacitance: redistribution of the venous reservoir as a cause of decompensation. *Circ Heart Fail*. 2011;4:669–675.
 72. Felker GM, Lee KL, Bull DA, et al. Diuretic strategies in patients with acute decompensated heart failure. *N Engl J Med*. 2011;364:797–805.
 73. Costanzo MR, Saltzberg M, O’Sullivan J, Sobotka P. Early ultrafiltration in patients with decompensated heart failure and diuretic resistance. *J Am Coll Cardiol*. 2005;46:2047–2051.
 74. Bartone C, Saghir S, Menon SG, et al. Comparison of ultrafiltration, nesiritide, and usual care in acute decompensated heart failure. *Congest Heart Fail*. 2008;14:298–301.
 75. Costanzo MR, Guglin ME, Saltzberg MT, et al. Ultrafiltration versus intravenous diuretics for patients hospitalized for acute decompensated heart failure. *J Am Coll Cardiol*. 2007;49:675–683.
 76. Bart BA, Goldsmith SR, Lee KL, et al. Ultrafiltration in decompensated heart failure with cardiorenal syndrome. *N Engl J Med*. 2012;367:2296–2304.
 77. Costanzo MR, Negoianu D, Jaski BE, et al. Aquapheresis versus intravenous diuretics and hospitalizations for heart failure. *JACC Heart Fail*. 2016;4:95–105.
 78. Koch M, Haastert B, Kohnle M, et al. Peritoneal dialysis relieves clinical symptoms and is well tolerated in patients with refractory heart failure and chronic kidney disease. *Eur J Heart Fail*. 2012;14:530–539.
 79. Bar-Zohar D, Sagie B, Lubezky N, Blum M, Klausner J, Abu-Abeid S. Laparoscopic implantation of the Tenckhoff catheter for the treatment of end-stage renal failure and congestive heart failure: experience with the pelvic fixation technique. *Isr Med Assoc J*. 2006;8:174–178.
 80. Ryckelynck JP, Lobbedez T, Valette B, et al. Peritoneal ultrafiltration and treatment-resistant heart failure. *Nephrol Dial Transplant*. 1998;13(suppl 4):56–59.
 81. Cove ME, MacLaren G, Federspiel WJ, Kellum JA. Bench to bedside review: extracorporeal carbon dioxide removal, past present and future. *Crit Care*. 2012;16:232.
 82. Hasin T, Topilsky Y, Schirger JA, et al. Changes in renal function after implantation of continuous-flow left ventricular assist devices. *J Am Coll Cardiol*. 2012;59:26–36.
 83. Karakala N, Juncos LA. How I treat: providing Continuous renal replacement therapy in patients on extracorporeal membrane oxygenation. *Clin J Amer Soc Nephrol*. 2020.
 84. Patel AM, Eduardo Rame J, Rudnick MR. How does the nephrologist manage an LVAD patient on chronic maintenance dialysis? *Semin Dial*. 2014;27:284–288.